Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Martin Land & Cattle

Physical address of dairy:

Number and Street	City	County	Zip Code
6100 Hinds RD	Oakdale	Stanislaus	95361

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1964

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X002-X012-X042-XXXX X187-X410-X005-XXXX

B. OPERATORS

Martin, Joe B

Operator name: Martin, Joe B

Telephone no.: (208) 847-2473 Landline, (209) 595-1900 Cellular

Mailing Address Number and Street	City	State	Zip Code
6100 Hinds RD	Oakdale	CA	95361

This operator is responsible for paying permit fees.

C. OWNERS

Martin, Joe B

Legal owner name: Martin, Joe B

Telephone no.: (208) 847-2473 Landline, (209) 595-1900 Cellular

Mailing Address Number and Street	City	State	Zip Code
6100 Hinds RD	Oakdale	CA	95361

This owner is responsible for paying permit fees.

Martin Trust, August & Elsie B

Legal owner name:	Martin Trust, August & Elsie B	Telephone no.:	(209) 838-1174	
			Landline	Cellular
7532 S Henry RD		Farmington	CA	95230
Mailing Address Number and Street		City	State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	385	165	300	175	0
Number under roof	690	0	0	0	0	0
Maximum number	690	400	200	400	175	0
Average number	690	385	165	300	175	0
Avg live weight (lbs)	1,400	1,450	800	600		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,882.34 *tons per reporting period*

Total nitrogen from manure: 349,451.88 *lbs per reporting period*

After ammonia losses (30% loss applied): 244,616.32 *lbs per reporting period*

Total phosphorus from manure: 55,578.22 *lbs per reporting period*

Total potassium from manure: 129,874.72 *lbs per reporting period*

Total salt from manure: 413,417.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 17,117,703 *gallons*

Total nitrogen generated: 72,200.64 *lbs*

Total phosphorus generated: 8,932.50 *lbs*

Total potassium generated: 118,429.94 *lbs*

Total salt generated: 890,288.13 *lbs*

	17,117,703	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	17,117,703	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/01/2023	Solid commercial fertilizer *CAN 17*	12.50 *ton*	As-is	0.1	17.000000	0.000000	0.000000	0.000000
05/01/2023	Solid commercial fertilizer *Ammonium Sulfate*	9.78 *ton*	Dry-weight	0.1	21.000000	0.000000	0.000000	0.000000
06/01/2023	Solid commercial fertilizer *UN32*	115.50 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	82,199.57	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	82,199.57	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
"17"	14	14	2	manure	X187-X041-X006-XXXX
"25"	21	21	2	manure	X187-X041-X006-XXXX
"5" Acres	5	5	0	none	X187-X041-X005-XXXX
"Dry"	6	6	0	none	X187-X041-X006-XXXX
Alf 1	10	10	2	manure	X187-X041-X006-XXXX
Alf 2	13	13	2	process wastewater	X187-X041-X006-XXXX
Alf 3	19	19	2	process wastewater	X187-X041-X006-XXXX
Alf 4	4	4	0	none	X187-X041-X006-XXXX
Alm 06	120	120	1	none	X187-X041-X003-XXXX
Alm 13	113	113	1	none	X187-X041-X003-XXXX
Delyo	18	18	2	manure	X002-X012-X061-XXXX
Dry Cows	31	31	2	manure	X187-X041-X006-XXXX
H1	30	30	2	both	X002-X012-X042-X000
H2	51	51	2	process wastewater	X002-X012-X042-X000
H3	49	49	2	process wastewater	X002-X012-X042-X000
H4	13	13	2	both	X002-X012-X042-X000
H5	16	16	2	both	X002-X012-X042-X000
H6	20	20	2	both	X002-X012-X042-X000
H7	34	34	2	manure	X002-X012-X042-X000
H8	11	11	2	manure	X002-X012-X042-X000
Pasture	16	16	2	manure	X187-X041-X006-XXXX
The Entrance	33	33	2	manure	X187-X041-X006-XXXX
Wal 07	50	50	1	none	X187-X042-X014-X000
Wal 15	87	87	1	none	X187-X041-X003-XXXX
Totals for areas that were used for application	399	399	34		
Totals for areas that were not used for application	385	385	4		
Land application area totals	784	784	38		

B. CROPS AND HARVESTS

"17"

Field name: "17"

"17"

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	246.40 *ton*	As-is		78.6	5,600.00	1,100.00	15,300.00		14.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.60	197.12	38.72	538.56	1,084.72

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	403.20 *ton*	As-is		62.2	4,200.00	500.00	4,500.00		4.07

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.80	241.92	28.80	259.20	886.15

"25"

Field name: "25"

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	363.13 *ton*	As-is		78.6	5,600.00	1,100.00	15,300.00		14.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.29	193.67	38.04	529.13	1,065.73

"25"

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	571.20 *ton*	As-is		62.8	4,700.00	500.00	4,200.00		4.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.20	255.68	27.20	228.48	975.41

Alf 1

Field name: Alf 1

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	167.00 *ton*	As-is		76.6	4,800.00	1,100.00	14,800.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.70	160.32	36.74	494.32	1,008.21

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	309.00 *ton*	As-is		65.8	4,000.00	500.00	4,000.00		5.51

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.90	247.20	30.90	247.20	1,164.57

Alf 2

Field name: Alf 2

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	192.50 *ton*	As-is		76.6	4,800.00	1,100.00	14,800.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.81	142.15	32.58	438.31	893.97

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	371.80 *ton*	As-is		65.0	3,800.00	500.00	4,100.00		4.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.60	217.36	28.60	234.52	970.97

Alf 3

Field name: Alf 3

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	307.80 *ton*	As-is		76.6	4,800.00	1,100.00	14,800.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.20	155.52	35.64	479.52	978.03

Alf 3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	587.10 *ton*	As-is		65.0	3,800.00	500.00	4,100.00		4.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.90	234.84	30.90	253.38	1,049.06

Alm 06

Field name: Alm 06

01/01/2023: Almond, in shell

Crop: Almond, in shell Acres planted: 120 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	450.00 *ton*	As-is		9.4	25,600.00	1,300.00	21,300.00		7.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	3.75	192.00	9.75	159.75	541.56

Alm 13

Field name: Alm 13

Alm 13

01/01/2023: Almond, in shell

Crop: Almond, in shell Acres planted: 113 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	440.70 *ton*	As-is		9.4	25,600.00	1,300.00	21,300.00		7.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	3.90	199.68	10.14	166.14	563.22

Delyo

Field name: Delyo

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	320.40 *ton*	As-is		77.8	4,100.00	1,000.00	13,100.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.80	145.96	35.60	466.36	916.77

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	480.60 *ton*	As-is		62.2	4,200.00	500.00	4,600.00		6.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.70	224.28	26.70	245.64	1,380.67

Dry Cows

Field name: Dry Cows

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	523.90 *ton*	As-is		77.6	5,900.00	1,200.00	15,700.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.90	199.42	40.56	530.66	1,105.40

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	799.80 *ton*	As-is		59.8	4,700.00	600.00	5,000.00		5.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.80	242.52	30.96	258.00	1,128.43

H1

Field name: H1

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	456.00 *ton*	As-is		67.2	5,200.00	800.00	9,200.00		9.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.20	158.08	24.32	279.68	913.36

H1

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	876.00 *ton*	As-is		64.4	4,700.00	500.00	4,800.00		6.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.20	274.48	29.20	280.32	1,307.72

H2

Field name: H2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 51 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	805.80 *ton*	As-is		67.2	5,200.00	800.00	9,200.00		9.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	164.32	25.28	290.72	949.42

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,591.20 *ton*	As-is		65.6	4,300.00	500.00	4,900.00		6.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.20	268.32	31.20	305.76	1,320.13

H3

Field name: H3

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	788.90 *ton*	As-is		63.6	5,800.00	900.00	10,500.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.10	186.76	28.98	338.10	1,277.57

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,367.10 *ton*	As-is		65.6	4,300.00	500.00	4,900.00		6.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.90	239.94	27.90	273.42	1,180.50

H4

Field name: H4

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	223.60 *ton*	As-is		63.6	5,800.00	900.00	10,500.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.20	199.52	30.96	361.20	1,364.85

H4

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	417.30 *ton*	As-is		64.4	4,200.00	600.00	4,600.00		5.87

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.10	269.64	38.52	295.32	1,341.60

H5

Field name: H5

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	264.00 *ton*	As-is		63.6	5,800.00	900.00	10,500.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	191.40	29.70	346.50	1,309.31

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	456.00 *ton*	As-is		63.4	4,300.00	600.00	4,200.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.50	245.10	34.20	239.40	1,063.96

H6

Field name: H6

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	344.00 *ton*	As-is		74.2	4,400.00	900.00	10,100.00		5.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.20	151.36	30.96	347.44	469.50

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	574.00 *ton*	As-is		64.0	4,200.00	550.00	4,900.00		5.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.70	241.08	31.57	281.26	1,159.25

H7

Field name: H7

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	540.60 *ton*	As-is		74.2	4,400.00	900.00	10,100.00		5.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.90	139.92	28.62	321.18	432.37

H7

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,006.40 *ton*	As-is		67.8	3,600.00	400.00	4,800.00		6.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.60	213.12	23.68	284.16	1,187.59

H8

Field name: H8

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	192.50 *ton*	As-is		74.2	4,400.00	900.00	10,100.00		5.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.50	154.00	31.50	353.50	475.88

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	303.60 *ton*	As-is		65.2	4,200.00	500.00	4,200.00		5.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.60	231.84	27.60	231.84	1,106.47

Pasture

Field name: Pasture

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	262.40 *ton*	As-is		67.7	5,000.00	900.00	7,100.00		9.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.40	164.00	29.52	232.88	1,021.30

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	393.60 *ton*	As-is		63.8	4,200.00	400.00	4,200.00		4.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.60	206.64	19.68	206.64	778.31

The Entrance

Field name: The Entrance

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	570.90 *ton*	As-is		67.7	5,000.00	900.00	7,100.00		9.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.30	173.00	31.14	245.66	1,077.35

The Entrance

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	854.70 *ton*	As-is		65.6	3,900.00	400.00	4,600.00		6.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.90	202.02	20.72	238.28	1,177.85

Wal 07

Field name: Wal 07

01/01/2023: Walnuts

Crop: Walnuts Acres planted: 50 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	120.00 *ton*	As-is		7.0	38,600.00	4,400.00	5,700.00		1.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.50	100.00	7.50	72.00	0.00
Total actual harvest content	2.40	185.28	21.12	27.36	79.46

Wal 15

Field name: Wal 15

Wal 15

01/01/2023: Walnuts

Crop: Walnuts Acres planted: 87 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	208.80 *ton*	As-is		7.0	38,600.00	4,400.00	5,700.00		1.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.50	100.00	7.50	72.00	0.00
Total actual harvest content	2.40	185.28	21.12	27.36	79.46

NUTRIENT BUDGET

A. LAND APPLICATIONS

"17" - 10/26/2022: Oats, silage-soft dough

Field name: "17"

Crop: Oats, silage-soft dough — Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	154.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	72.94	2,353,120.00 *gal*
Application event totals		1.42	0.00	0.00	72.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	55.88	1,802,836.00 *gal*
Application event totals		1.09	0.00	0.00	55.88	

"17" - 06/02/2023: Corn, silage

Field name: "17"

Crop: Corn, silage — Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	140.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

"17" - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	75.17	2,425,150.00 *gal*
Application event totals		1.46	0.00	0.00	75.17	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.41	0.00	0.00	72.74	2,346,624.00 *gal*
Application event totals		65.41	0.00	0.00	72.74	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.75	2,282,420.00 *gal*
Application event totals		1.37	0.00	0.00	70.75	

08/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.42	2,239,566.00 *gal*
Application event totals		1.35	0.00	0.00	69.42	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	66.92	2,158,982.00 *gal*
Application event totals		1.30	0.00	0.00	66.92	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.99	2,096,612.00 *gal*
Application event totals		1.26	0.00	0.00	64.99	

"25" - 10/26/2022: Oats, silage-soft dough

Field name: "25"

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	231.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.88	3,623,592.00 *gal*
Application event totals		1.45	0.00	0.00	74.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.11	0.00	0.00	56.93	2,755,242.00 *gal*
Application event totals		1.11	0.00	0.00	56.93	

"25" - 06/02/2023: Corn, silage

Field name: "25"

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	210.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

"25" - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.44	0.00	0.00	73.95	3,578,715.00 *gal*
Application event totals		1.44	0.00	0.00	73.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.41	0.00	0.00	72.84	3,525,039.00 *gal*
Application event totals		65.41	0.00	0.00	72.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	70.83	3,427,872.00 *gal*
Application event totals		1.38	0.00	0.00	70.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.61	3,320,499.00 *gal*
Application event totals		1.33	0.00	0.00	68.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	67.27	3,255,420.00 *gal*
Application event totals		1.31	0.00	0.00	67.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	65.40	3,165,015.00 *gal*
Application event totals		1.27	0.00	0.00	65.40	

Alf 1 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 1

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	110.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	73.21	1,687,211.00 *gal*
Application event totals		1.42	0.00	0.00	73.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	56.72	1,307,013.00 *gal*
Application event totals		1.10	0.00	0.00	56.72	

Alf 1 - 06/01/2023: Corn, silage

Field name: Alf 1

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	100.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

Alf 1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.52	0.00	0.00	78.20	1,802,150.00 *gal*
Application event totals		1.52	0.00	0.00	78.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.46	0.00	0.00	75.02	1,728,916.00 *gal*
Application event totals		65.46	0.00	0.00	75.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	71.28	1,642,630.00 *gal*
Application event totals		1.38	0.00	0.00	71.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.65	1,581,920.00 *gal*
Application event totals		1.33	0.00	0.00	68.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	67.68	1,559,640.00 *gal*
Application event totals		1.31	0.00	0.00	67.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.75	1,492,113.00 *gal*
Application event totals		1.26	0.00	0.00	64.75	

Alf 2 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 2

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.53	7.41	210.34	823.95	225,980.00 *gal*
Canal	Surface water	1.35	0.00	0.00	69.52	2,082,795.00 *gal*
Application event totals		45.88	7.41	210.34	893.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.76	7.04	113.46	1,645.78	323,715.00 *gal*
Application event totals		62.76	7.04	113.46	1,645.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.19	6.53	105.20	1,526.00	300,155.00 *gal*
Application event totals		58.19	6.53	105.20	1,526.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.37	4.38	106.22	832.63	167,150.00 *gal*
Canal	Surface water	1.08	0.00	0.00	55.69	1,668,238.00 *gal*
Application event totals		34.45	4.38	106.22	888.31	

Alf 2 - 06/01/2023: Corn, silage

Field name: Alf 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Alf 2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.39	8.18	198.59	1,556.66	312,500.00 *gal*
Canal	Surface water	1.43	0.00	0.00	73.73	2,208,674.00 *gal*
Application event totals		63.82	8.18	198.59	1,630.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.39	0.00	0.00	71.51	2,142,205.00 *gal*
Application event totals		65.39	0.00	0.00	71.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.02	7.22	175.13	1,372.77	275,584.00 *gal*
Canal	Surface water	1.35	0.00	0.00	69.74	2,089,295.00 *gal*
Application event totals		56.37	7.22	175.13	1,442.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.73	5.34	129.64	1,016.17	203,995.00 *gal*
Canal	Surface water	1.34	0.00	0.00	68.89	2,063,958.00 *gal*
Application event totals		42.06	5.34	129.64	1,085.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.39	7.36	87.08	589.08	200,366.00 *gal*
Canal	Surface water	1.30	0.00	0.00	66.97	2,006,238.00 *gal*
Application event totals		59.69	7.36	87.08	656.05	

Alf 2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.99	1,947,114.00 *gal*
Application event totals		1.26	0.00	0.00	64.99	

Alf 3 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 3

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.44	9.39	266.56	1,044.17	418,552.00 *gal*
Canal	Surface water	1.45	0.00	0.00	74.72	3,271,553.00 *gal*
Application event totals		57.89	9.39	266.56	1,118.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.03	7.52	121.18	1,757.80	505,325.00 *gal*
Application event totals		67.03	7.52	121.18	1,757.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.40	5.77	92.94	1,348.10	387,546.00 *gal*
Application event totals		51.40	5.77	92.94	1,348.10	

Alf 3 - 10/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.62	4.02	97.48	764.09	224,187.00 *gal*
Canal	Surface water	1.10	0.00	0.00	56.88	2,490,672.00 *gal*
Application event totals		31.73	4.02	97.48	820.97	

Alf 3 - 06/01/2023: Corn, silage

Field name: Alf 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.74	9.41	228.36	1,789.94	525,175.00 *gal*
Canal	Surface water	1.42	0.00	0.00	73.24	3,206,782.00 *gal*
Application event totals		73.16	9.41	228.36	1,863.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.38	0.00	0.00	71.04	3,110,585.00 *gal*
Application event totals		65.38	0.00	0.00	71.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	68.79	3,011,975.00 *gal*
Application event totals		1.34	0.00	0.00	68.79	

Alf 3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.33	8.18	198.42	1,555.28	456,325.00 *gal*
Canal	Surface water	1.30	0.00	0.00	67.16	2,940,706.00 *gal*
Application event totals		63.64	8.18	198.42	1,622.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.98	5.38	130.45	1,022.48	300,000.00 *gal*
Canal	Surface water	1.27	0.00	0.00	65.20	2,854,854.00 *gal*
Application event totals		42.24	5.38	130.45	1,087.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.26	5.70	67.49	456.60	226,985.00 *gal*
Canal	Surface water	1.25	0.00	0.00	64.53	2,825,585.00 *gal*
Application event totals		46.51	5.70	67.49	521.13	

Alm 06 - 01/01/2023: Almond, in shell

Field name: Alm 06

Crop: Almond, in shell

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Application event totals		42.50	0.00	0.00	0.00	

Alm 06 - 01/01/2023: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Application event totals		42.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Irrigation Well	Ground water	2.80	0.00	0.00	229.56	22,765,800.00 *gal*
Application event totals		45.30	0.00	0.00	229.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ammonium Sulfate	Solid commercial fertilizer	46.20	0.00	0.00	0.00	
Irrigation Well	Ground water	2.60	0.00	0.00	213.05	21,128,280.00 *gal*
Application event totals		48.80	0.00	0.00	213.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.50	0.00	0.00	205.11	20,341,440.00 *gal*
Application event totals		2.50	0.00	0.00	205.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.40	0.00	0.00	196.22	19,458,960.00 *gal*
Application event totals		2.40	0.00	0.00	196.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.35	0.00	0.00	192.34	19,075,080.00 *gal*
Application event totals		2.35	0.00	0.00	192.34	

Alm 06 - 01/01/2023: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.24	0.00	0.00	183.78	18,225,360.00 *gal*
Application event totals		2.24	0.00	0.00	183.78	

Alm 13 - 01/01/2023: Almond, in shell

Field name: Alm 13

Crop: Almond, in shell

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Application event totals		42.50	0.00	0.00	0.00	

03/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Application event totals		42.50	0.00	0.00	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Irrigation Well	Ground water	2.60	0.00	0.00	226.33	20,431,304.00 *gal*
Application event totals		45.10	0.00	0.00	226.33	

Alm 13 - 01/01/2023: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN 17	Solid commercial fertilizer	46.20	0.00	0.00	0.00	
Canal	Surface water	1.50	0.00	0.00	77.03	20,058,856.00 *gal*
Application event totals		47.70	0.00	0.00	77.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.44	0.00	0.00	212.57	19,189,095.00 *gal*
Application event totals		2.44	0.00	0.00	212.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.37	0.00	0.00	207.03	18,688,957.00 *gal*
Application event totals		2.37	0.00	0.00	207.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.31	0.00	0.00	201.40	18,181,135.00 *gal*
Application event totals		2.31	0.00	0.00	201.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.23	0.00	0.00	194.54	17,561,556.00 *gal*
Application event totals		2.23	0.00	0.00	194.54	

Delyo - 11/05/2022: Oats, silage-soft dough

Field name: Delyo

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Delyo - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Separator solids	202.08	65.70	64.86	2,694.38	252.00 *ton*
Application event totals		202.08	65.70	64.86	2,694.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.57	2,927,070.00 *gal*
Application event totals		1.37	0.00	0.00	70.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	56.50	2,343,600.00 *gal*
Application event totals		1.10	0.00	0.00	56.50	

Delyo - 06/11/2023: Corn, silage

Field name: Delyo

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.64	82.58	263.30	7,707.39	198.00 *ton*
Application event totals		210.64	82.58	263.30	7,707.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.43	0.00	0.00	73.87	3,064,050.00 *gal*
Application event totals		1.43	0.00	0.00	73.87	

Delyo - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.40	0.00	0.00	72.26	2,997,414.00 *gal*
Application event totals		65.40	0.00	0.00	72.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	69.81	2,895,750.00 *gal*
Application event totals		1.36	0.00	0.00	69.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	68.94	2,859,750.00 *gal*
Application event totals		1.34	0.00	0.00	68.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.29	0.00	0.00	66.41	2,754,540.00 *gal*
Application event totals		1.29	0.00	0.00	66.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	65.19	2,704,050.00 *gal*
Application event totals		1.27	0.00	0.00	65.19	

Dry Cows - 10/26/2022: Oats, silage-soft dough

Field name: Dry Cows

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dry Cows - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	341.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.41	0.00	0.00	72.65	5,189,865.00 *gal*
Application event totals		1.41	0.00	0.00	72.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	55.38	3,955,972.00 *gal*
Application event totals		1.08	0.00	0.00	55.38	

Dry Cows - 06/02/2023: Corn, silage

Field name: Dry Cows

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	310.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	75.30	5,379,306.00 *gal*
Application event totals		1.46	0.00	0.00	75.30	

Dry Cows - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.40	0.00	0.00	72.18	5,156,106.00 *gal*
Application event totals		65.40	0.00	0.00	72.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.42	5,030,897.00 *gal*
Application event totals		1.37	0.00	0.00	70.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.32	0.00	0.00	67.83	4,845,672.00 *gal*
Application event totals		1.32	0.00	0.00	67.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	67.21	4,801,125.00 *gal*
Application event totals		1.31	0.00	0.00	67.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	64.51	4,608,522.00 *gal*
Application event totals		1.25	0.00	0.00	64.51	

H1 - 11/03/2022: Oats, silage-soft dough

Field name: H1

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

H1 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	202.08	65.70	64.86	2,694.38	420.00 *ton*
Application event totals		202.08	65.70	64.86	2,694.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.40	0.00	0.00	72.21	4,992,450.00 *gal*
Application event totals		1.40	0.00	0.00	72.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	56.42	3,900,240.00 *gal*
Application event totals		1.10	0.00	0.00	56.42	

H1 - 06/09/2023: Corn, silage

Field name: H1

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.72	12.13	161.40	826.91	754,500.00 *gal*
Canal	Surface water	1.44	0.00	0.00	74.31	5,137,650.00 *gal*
Application event totals		93.16	12.13	161.40	901.23	

H1 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.31	0.00	0.00	67.30	4,652,970.00 *gal*
Application event totals		65.31	0.00	0.00	67.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.14	11.55	172.72	1,192.35	500,755.00 *gal*
Canal	Surface water	1.20	0.00	0.00	61.76	4,270,050.00 *gal*
Application event totals		135.34	11.55	172.72	1,254.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.39	0.00	0.00	71.35	4,932,610.00 *gal*
Application event totals		1.39	0.00	0.00	71.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.50	4,735,450.00 *gal*
Application event totals		1.33	0.00	0.00	68.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	52.94	3,659,910.00 *gal*
Application event totals		1.03	0.00	0.00	52.94	

H2 - 11/04/2022: Oats, silage-soft dough

Field name: H2

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

H2 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.13	11.68	151.08	974.54	1,575,615.00 *gal*
Canal	Surface water	1.42	0.00	0.00	73.30	8,614,665.00 *gal*
Application event totals		96.56	11.68	151.08	1,047.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.64	16.07	107.11	1,049.70	1,087,321.00 *gal*
Application event totals		86.64	16.07	107.11	1,049.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.08	6,590,373.00 *gal*
Application event totals		1.09	0.00	0.00	56.08	

H2 - 06/09/2023: Corn, silage

Field name: H2

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	75.07	8,822,439.00 *gal*
Application event totals		1.46	0.00	0.00	75.07	

H2 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.35	0.00	0.00	69.52	8,170,965.00 *gal*
Application event totals		65.35	0.00	0.00	69.52	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.35	8,033,265.00 *gal*
Application event totals		1.33	0.00	0.00	68.35	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.23	11.90	177.99	1,228.70	877,233.00 *gal*
Canal	Surface water	1.26	0.00	0.00	64.94	7,632,405.00 *gal*
Application event totals		139.49	11.90	177.99	1,293.64	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.31	8.20	122.72	847.19	604,855.00 *gal*
Canal	Surface water	1.20	0.00	0.00	61.76	7,257,963.00 *gal*
Application event totals		96.51	8.20	122.72	908.95	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.04	0.00	0.00	53.73	6,314,565.00 *gal*
Application event totals		1.04	0.00	0.00	53.73	

H3 - 11/03/2022: Oats, silage-soft dough

Field name: H3

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

H3 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.43	0.00	0.00	73.77	8,329,902.00 *gal*
Application event totals		1.43	0.00	0.00	73.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.12	9.47	122.47	789.99	1,227,150.00 *gal*
Application event totals		77.12	9.47	122.47	789.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.05	21.33	142.21	1,393.79	1,387,123.00 *gal*
Canal	Surface water	1.13	0.00	0.00	57.97	6,546,302.00 *gal*
Application event totals		116.17	21.33	142.21	1,451.76	

H3 - 06/10/2023: Corn, silage

Field name: H3

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.02	10.58	140.81	721.45	1,075,180.00 *gal*
Canal	Surface water	1.43	0.00	0.00	73.48	8,296,974.00 *gal*
Application event totals		81.45	10.58	140.81	794.93	

H3 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.33	0.00	0.00	68.51	7,736,463.00 *gal*
Application event totals		65.33	0.00	0.00	68.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	145.92	12.56	187.90	1,297.10	889,751.00 *gal*
Canal	Surface water	1.27	0.00	0.00	65.63	7,410,417.00 *gal*
Application event totals		147.20	12.56	187.90	1,362.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	64.45	7,277,284.00 *gal*
Application event totals		1.25	0.00	0.00	64.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	61.31	6,923,112.00 *gal*
Application event totals		1.19	0.00	0.00	61.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.16	0.00	0.00	59.52	6,720,448.00 *gal*
Application event totals		1.16	0.00	0.00	59.52	

H4 - 11/04/2022: Oats, silage-soft dough

Field name: H4

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

H4 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	70.95	2,125,656.00 *gal*
Application event totals		1.38	0.00	0.00	70.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.57	13.09	87.24	855.01	225,755.00 *gal*
Application event totals		70.57	13.09	87.24	855.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	136.97	25.40	169.32	1,659.44	438,155.00 *gal*
Application event totals		136.97	25.40	169.32	1,659.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	56.89	1,704,287.00 *gal*
Application event totals		1.10	0.00	0.00	56.89	

H4 - 06/11/2023: Corn, silage

Field name: H4

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.64	82.58	263.30	7,707.39	143.00 *ton*
Application event totals		210.64	82.58	263.30	7,707.39	

H4 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.50	0.00	0.00	77.03	2,307,656.00 *gal*
Application event totals		1.50	0.00	0.00	77.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.41	0.00	0.00	72.65	2,176,356.00 *gal*
Application event totals		65.41	0.00	0.00	72.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	68.76	2,059,915.00 *gal*
Application event totals		1.34	0.00	0.00	68.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.28	0.00	0.00	66.01	1,977,430.00 *gal*
Application event totals		1.28	0.00	0.00	66.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.65	1,936,857.00 *gal*
Application event totals		1.26	0.00	0.00	64.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.23	0.00	0.00	63.36	1,898,117.00 *gal*
Application event totals		1.23	0.00	0.00	63.36	

H5 - 11/04/2022: Oats, silage-soft dough

Field name: H5

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.66	9.17	118.57	764.86	387,955.00 *gal*
Canal	Surface water	1.41	0.00	0.00	72.34	2,667,440.00 *gal*
Application event totals		76.07	9.17	118.57	837.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.79	13.31	88.75	869.79	282,656.00 *gal*
Application event totals		71.79	13.31	88.75	869.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.00	2,064,880.00 *gal*
Application event totals		1.09	0.00	0.00	56.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.37	9.52	63.49	622.29	202,224.00 *gal*
Application event totals		51.37	9.52	63.49	622.29	

H5 - 06/10/2023: Corn, silage

Field name: H5

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

H5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.64	82.58	263.30	7,707.39	176.00 *ton*
Application event totals		210.64	82.58	263.30	7,707.39	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.69	2,753,840.00 *gal*
Application event totals		1.45	0.00	0.00	74.69	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.41	0.00	0.00	72.65	2,678,640.00 *gal*
Application event totals		65.41	0.00	0.00	72.65	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.48	2,562,000.00 *gal*
Application event totals		1.35	0.00	0.00	69.48	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	67.61	2,493,040.00 *gal*
Application event totals		1.31	0.00	0.00	67.61	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	64.49	2,377,840.00 *gal*
Application event totals		1.25	0.00	0.00	64.49	

H5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	61.30	2,260,080.00 *gal*
Application event totals		1.19	0.00	0.00	61.30	

H6 - 11/05/2022: Oats, silage-soft dough

Field name: H6

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	202.08	65.70	64.86	2,694.38	280.00 *ton*
Application event totals		202.08	65.70	64.86	2,694.38	

10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.70	3,443,080.00 *gal*
Application event totals		1.45	0.00	0.00	74.70	

04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	56.85	2,620,040.00 *gal*
Application event totals		1.10	0.00	0.00	56.85	

H6 - 06/11/2023: Corn, silage

Field name: H6

Crop: Corn, silage

Plant date: 06/11/2023

H6 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.43	0.00	0.00	73.67	3,395,500.00 *gal*
Application event totals		1.43	0.00	0.00	73.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.36	0.00	0.00	70.24	3,237,500.00 *gal*
Application event totals		65.36	0.00	0.00	70.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.40	3,152,660.00 *gal*
Application event totals		1.33	0.00	0.00	68.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.70	7.03	105.20	726.20	203,323.00 *gal*
Canal	Surface water	1.26	0.00	0.00	65.12	3,001,500.00 *gal*
Application event totals		82.96	7.03	105.20	791.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.47	11.92	178.30	1,230.84	344,612.00 *gal*
Canal	Surface water	1.25	0.00	0.00	64.10	2,954,500.00 *gal*
Application event totals		139.71	11.92	178.30	1,294.94	

H6 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.18	0.00	0.00	60.58	2,791,900.00 *gal*
Application event totals		1.18	0.00	0.00	60.58	

H7 - 11/05/2022: Oats, silage-soft dough

Field name: H7

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	202.08	65.70	64.86	2,694.38	476.00 *ton*
Application event totals		202.08	65.70	64.86	2,694.38	

10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.40	0.00	0.00	72.28	5,662,904.00 *gal*
Application event totals		1.40	0.00	0.00	72.28	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.11	0.00	0.00	57.29	4,488,680.00 *gal*
Application event totals		1.11	0.00	0.00	57.29	

H7 - 06/11/2023: Corn, silage

Field name: H7

Crop: Corn, silage

Plant date: 06/11/2023

H7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.64	82.58	263.30	7,707.39	374.00 *ton*
Application event totals		210.64	82.58	263.30	7,707.39	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	75.34	5,903,284.00 *gal*
Application event totals		1.46	0.00	0.00	75.34	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.40	0.00	0.00	72.27	5,662,768.00 *gal*
Application event totals		65.40	0.00	0.00	72.27	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.57	5,451,084.00 *gal*
Application event totals		1.35	0.00	0.00	69.57	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.32	0.00	0.00	67.81	5,313,044.00 *gal*
Application event totals		1.32	0.00	0.00	67.81	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	65.09	5,099,932.00 *gal*
Application event totals		1.26	0.00	0.00	65.09	

H7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	61.28	4,801,650.00 *gal*
Application event totals		1.19	0.00	0.00	61.28	

H8 - 11/05/2022: Oats, silage-soft dough

Field name: H8

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	202.08	65.70	64.86	2,694.38	154.00 *ton*
Application event totals		202.08	65.70	64.86	2,694.38	

10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.40	0.00	0.00	72.27	1,832,072.00 *gal*
Application event totals		1.40	0.00	0.00	72.27	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.07	0.00	0.00	55.11	1,396,956.00 *gal*
Application event totals		1.07	0.00	0.00	55.11	

H8 - 06/11/2023: Corn, silage

Field name: H8

Crop: Corn, silage

Plant date: 06/11/2023

H8 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.64	82.58	263.30	7,707.39	121.00 *ton*
Application event totals		210.64	82.58	263.30	7,707.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	75.22	1,906,652.00 *gal*
Application event totals		1.46	0.00	0.00	75.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.42	0.00	0.00	73.00	1,850,398.00 *gal*
Application event totals		65.42	0.00	0.00	73.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.34	1,782,968.00 *gal*
Application event totals		1.37	0.00	0.00	70.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	68.44	1,734,975.00 *gal*
Application event totals		1.33	0.00	0.00	68.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.28	0.00	0.00	66.06	1,674,585.00 *gal*
Application event totals		1.28	0.00	0.00	66.06	

H8 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	64.22	1,627,978.00 *gal*
Application event totals		1.25	0.00	0.00	64.22	

Pasture - 10/27/2022: Oats, silage-soft dough

Field name: Pasture

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	176.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.41	0.00	0.00	72.68	2,679,792.00 *gal*
Application event totals		1.41	0.00	0.00	72.68	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.00	2,064,944.00 *gal*
Application event totals		1.09	0.00	0.00	56.00	

Pasture - 06/02/2023: Corn, silage

Field name: Pasture

Crop: Corn, silage

Plant date: 06/02/2023

Pasture - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	160.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.43	0.00	0.00	73.50	2,709,904.00 *gal*
Application event totals		1.43	0.00	0.00	73.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.39	0.00	0.00	71.52	2,637,072.00 *gal*
Application event totals		65.39	0.00	0.00	71.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.57	2,565,216.00 *gal*
Application event totals		1.35	0.00	0.00	69.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	68.83	2,537,968.00 *gal*
Application event totals		1.34	0.00	0.00	68.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	67.64	2,494,000.00 *gal*
Application event totals		1.31	0.00	0.00	67.64	

Pasture - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.94	2,394,544.00 *gal*
Application event totals		1.26	0.00	0.00	64.94	

The Entrance - 10/27/2022: Oats, silage-soft dough

Field name: The Entrance

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.69	55.65	403.47	2,316.07	363.00 *ton*
Application event totals		199.69	55.65	403.47	2,316.07	

10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.90	5,695,734.00 *gal*
Application event totals		1.45	0.00	0.00	74.90	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.37	4,286,634.00 *gal*
Application event totals		1.09	0.00	0.00	56.37	

The Entrance - 06/02/2023: Corn, silage

Field name: The Entrance

Crop: Corn, silage

Plant date: 06/02/2023

The Entrance - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.28	73.71	508.95	6,458.40	330.00 *ton*
Application event totals		215.28	73.71	508.95	6,458.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.44	0.00	0.00	74.30	5,650,425.00 *gal*
Application event totals		1.44	0.00	0.00	74.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Canal	Surface water	1.41	0.00	0.00	72.60	5,520,867.00 *gal*
Application event totals		65.41	0.00	0.00	72.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.66	5,373,324.00 *gal*
Application event totals		1.37	0.00	0.00	70.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	68.98	5,245,845.00 *gal*
Application event totals		1.34	0.00	0.00	68.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	66.92	5,089,458.00 *gal*
Application event totals		1.30	0.00	0.00	66.92	

The Entrance - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	65.19	4,957,458.00 *gal*
Application event totals		1.27	0.00	0.00	65.19	

Wal 07 - 01/01/2023: Walnuts

Field name: Wal 07

Crop: Walnuts

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Irrigation Well	Ground water	2.66	0.00	0.00	214.04	9,160,250.00 *gal*
Application event totals		50.66	0.00	0.00	214.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.56	0.00	0.00	206.23	8,826,250.00 *gal*
Application event totals		2.56	0.00	0.00	206.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Irrigation Well	Ground water	2.44	0.00	0.00	196.34	8,402,750.00 *gal*
Application event totals		37.64	0.00	0.00	196.34	

Wal 07 - 01/01/2023: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Irrigation Well	Ground water	2.33	0.00	0.00	187.18	8,010,650.00 *gal*
Application event totals		37.53	0.00	0.00	187.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.29	0.00	0.00	184.58	7,899,300.00 *gal*
Application event totals		2.29	0.00	0.00	184.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.17	0.00	0.00	174.82	7,481,600.00 *gal*
Application event totals		2.17	0.00	0.00	174.82	

Wal 15 - 01/01/2023: Walnuts

Field name: Wal 15

Crop: Walnuts

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Irrigation Well	Ground water	2.64	0.00	0.00	212.81	15,847,137.00 *gal*
Application event totals		50.64	0.00	0.00	212.81	

Wal 15 - 01/01/2023: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.50	0.00	0.00	200.96	14,965,044.00 *gal*
Application event totals		2.50	0.00	0.00	200.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Irrigation Well	Ground water	2.43	0.00	0.00	195.49	14,557,362.00 *gal*
Application event totals		37.63	0.00	0.00	195.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Irrigation Well	Ground water	2.28	0.00	0.00	183.57	13,669,875.00 *gal*
Application event totals		37.48	0.00	0.00	183.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.16	0.00	0.00	173.74	12,938,205.00 *gal*
Application event totals		2.16	0.00	0.00	173.74	

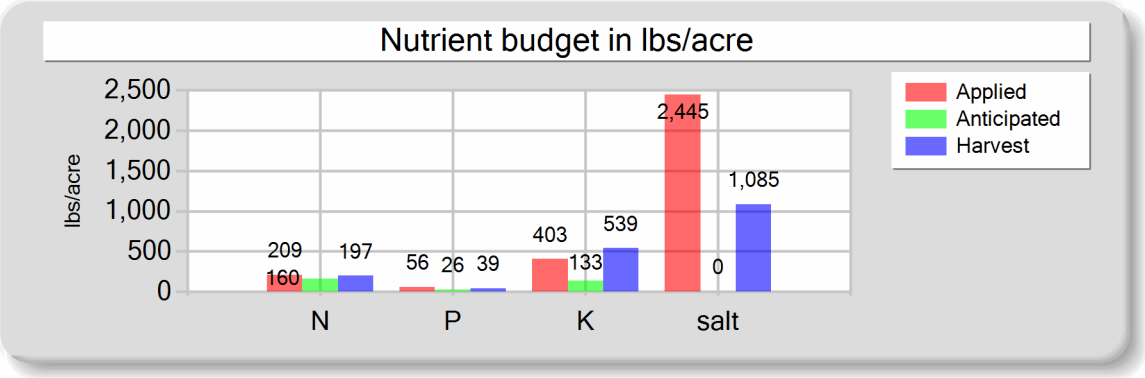
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.02	0.00	0.00	162.29	12,085,605.00 *gal*
Application event totals		2.02	0.00	0.00	162.29	

B. NUTRIENT BUDGET

"17" - 10/26/2022: Oats, silage-soft dough

Field name: "17" Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.50	0.00	0.00	128.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.19	55.65	403.47	2,444.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	197.12	38.72	538.56	1,084.72
Nutrient balance	12.07	16.93	-135.09	1,360.17
Applied to removed ratio	1.06	1.44	0.75	2.25

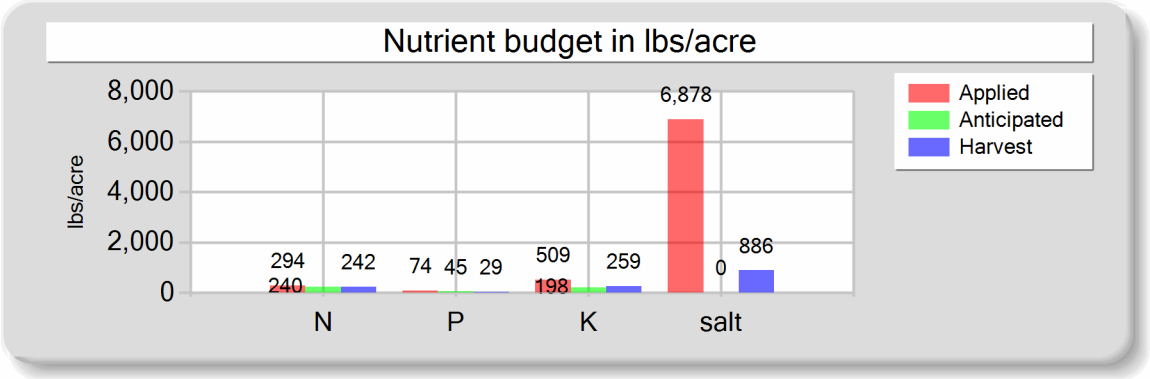
Fresh water applied	
4,155,956.00	*gallons*
153.05	*acre-inches*
10.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

"17" - 06/02/2023: Corn, silage

Field name: "17" Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.16	0.00	0.00	419.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.44	73.71	508.95	6,878.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.92	28.80	259.20	886.15
Nutrient balance	52.52	44.91	249.75	5,992.22
Applied to removed ratio	1.22	2.56	1.96	7.76

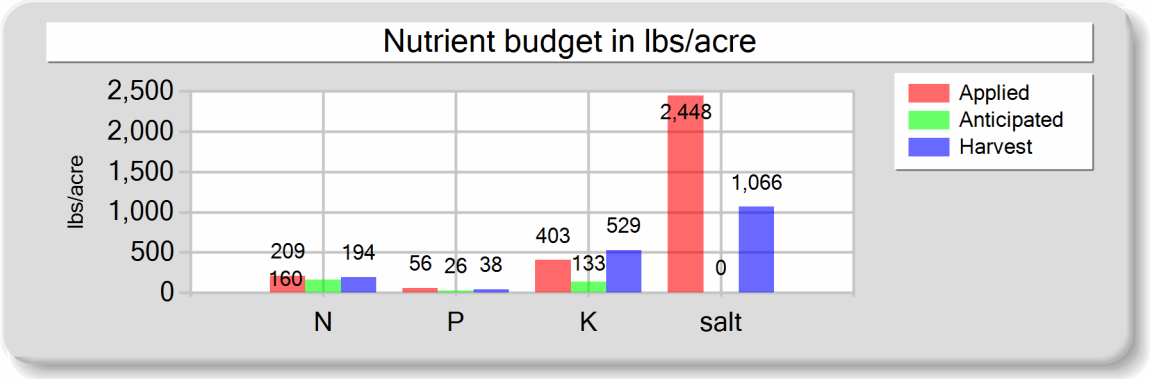
Fresh water applied	
13,549,354.00	*gallons*
498.98	*acre-inches*
35.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

"25" - 10/26/2022: Oats, silage-soft dough

Field name: "25" Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.56	0.00	0.00	131.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.25	55.65	403.47	2,447.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	193.67	38.04	529.13	1,065.73
Nutrient balance	15.58	17.61	-125.66	1,382.15
Applied to removed ratio	1.08	1.46	0.76	2.30

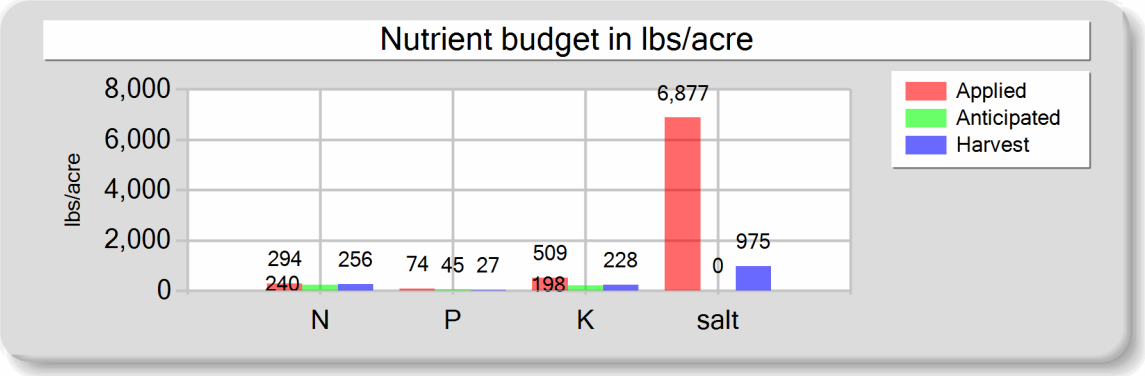
Fresh water applied	
6,378,834.00	*gallons*
234.91	*acre-inches*
11.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

"25" - 06/02/2023: Corn, silage

Field name: "25" Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.14	0.00	0.00	418.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.42	73.71	508.95	6,877.31
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	255.68	27.20	228.48	975.41
Nutrient balance	38.74	46.51	280.47	5,901.89
Applied to removed ratio	1.15	2.71	2.23	7.05

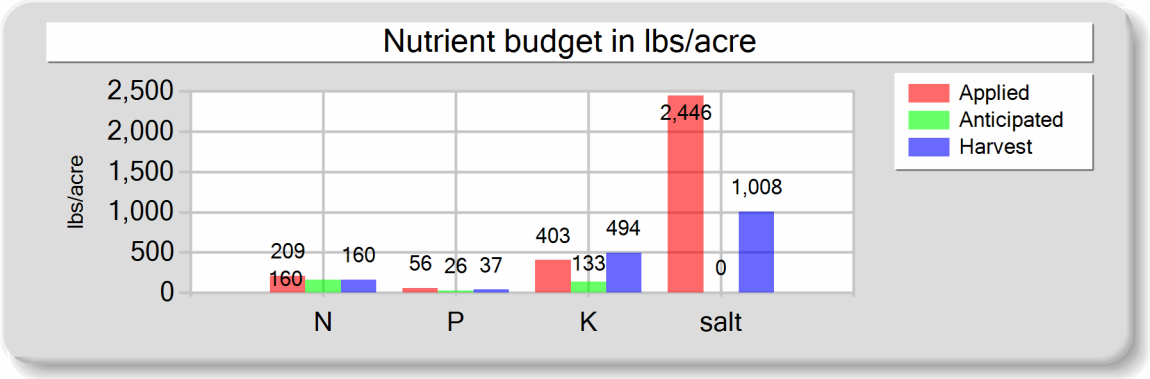
Fresh water applied	
20,272,560.00	*gallons*
746.57	*acre-inches*
35.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 1 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 1 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.52	0.00	0.00	129.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.21	55.65	403.47	2,446.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	160.32	36.74	494.32	1,008.21
Nutrient balance	48.89	18.91	-90.85	1,437.79
Applied to removed ratio	1.30	1.51	0.82	2.43

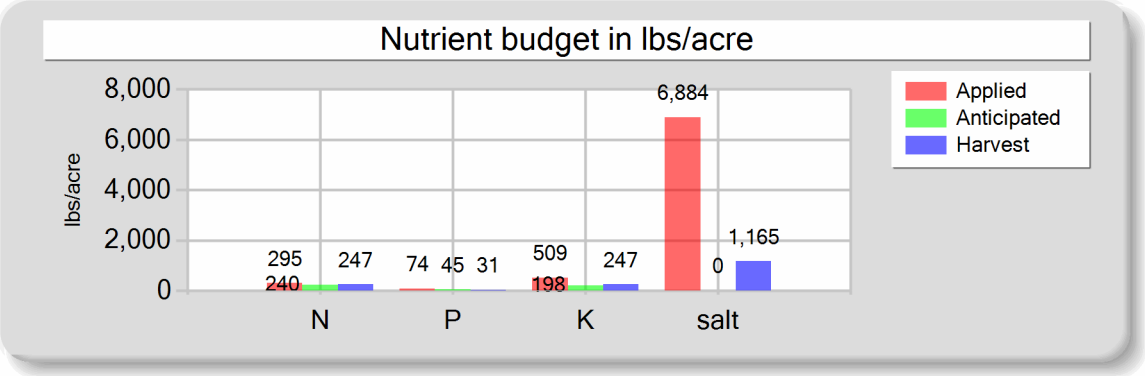
Fresh water applied	
2,994,224.00	*gallons*
110.27	*acre-inches*
11.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 1 - 06/01/2023: Corn, silage

Field name: Alf 1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.27	0.00	0.00	425.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.55	73.71	508.95	6,883.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.20	30.90	247.20	1,164.57
Nutrient balance	47.35	42.81	261.75	5,719.41
Applied to removed ratio	1.19	2.39	2.06	5.91

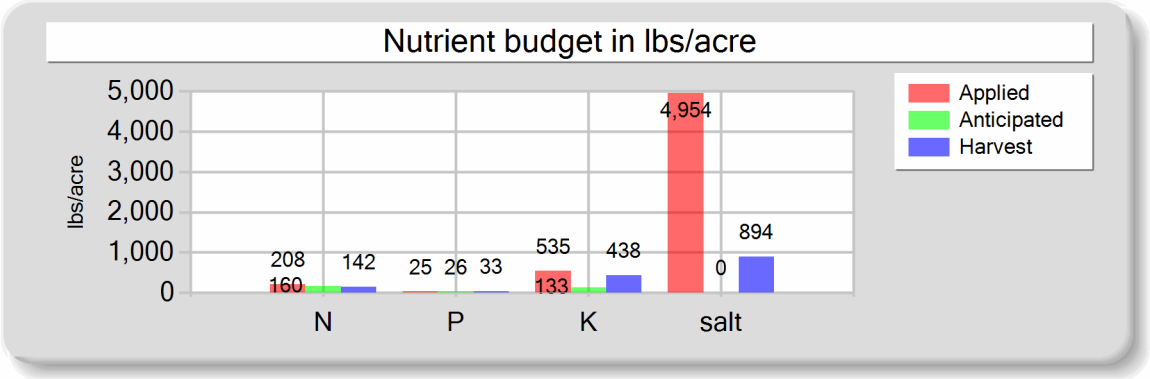
Fresh water applied	
9,807,369.00	*gallons*
361.17	*acre-inches*
36.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 2 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 2 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.85	25.37	535.22	4,828.35
Fresh water	2.43	0.00	0.00	125.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.28	25.37	535.22	4,953.56
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.15	32.58	438.31	893.97
Nutrient balance	66.13	-7.21	96.92	4,059.59
Applied to removed ratio	1.47	0.78	1.22	5.54

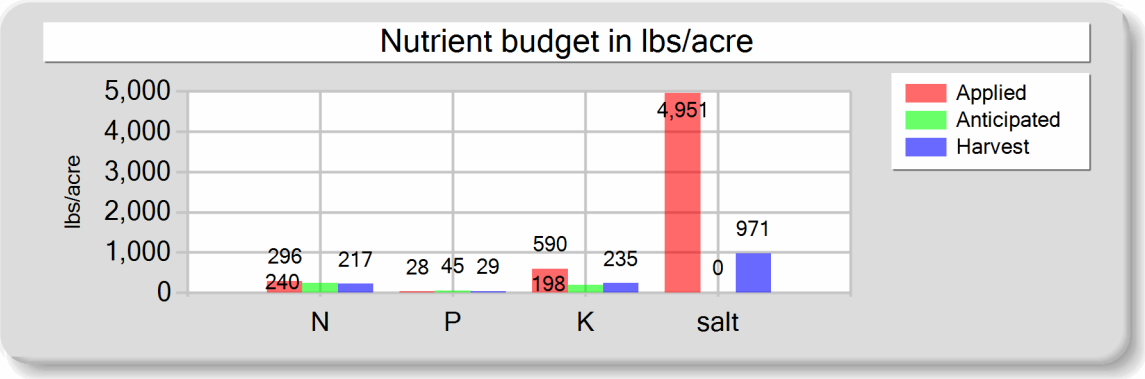
Fresh water applied	
3,751,033.00	*gallons*
138.14	*acre-inches*
10.63	*inches/acre*

Process wastewater applied	
1,017,000.00	*gallons*
37.45	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 2 - 06/01/2023: Corn, silage

Field name: Alf 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.52	28.10	590.44	4,534.68
Fresh water	8.08	0.00	0.00	415.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.60	28.10	590.44	4,950.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.36	28.60	234.52	970.97
Nutrient balance	78.24	-0.50	355.92	3,979.54
Applied to removed ratio	1.36	0.98	2.52	5.10

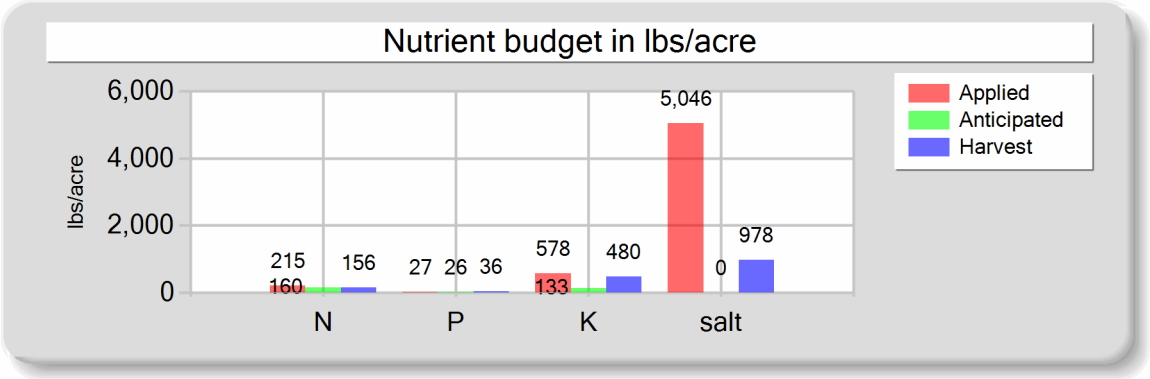
Fresh water applied	
12,457,484.00	*gallons*
458.77	*acre-inches*
35.29	*inches/acre*

Process wastewater applied	
992,445.00	*gallons*
36.55	*acre-inches*
2.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 3 - 10/25/2022: Oats, silage-soft dough

Field name: Alf 3 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.49	26.71	578.16	4,914.15
Fresh water	2.56	0.00	0.00	131.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.05	26.71	578.16	5,045.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	155.52	35.64	479.52	978.03
Nutrient balance	59.53	-8.93	98.64	4,067.73
Applied to removed ratio	1.38	0.75	1.21	5.16

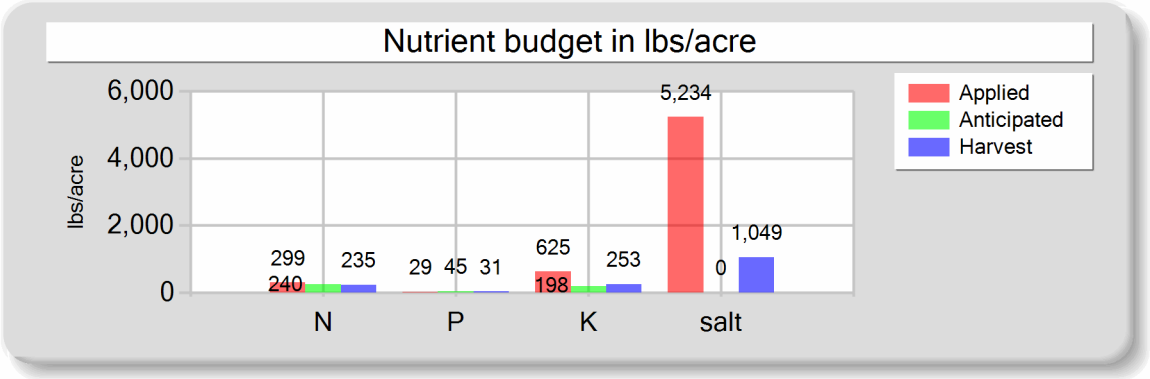
Fresh water applied	
5,762,225.00	*gallons*
212.20	*acre-inches*
11.17	*inches/acre*

Process wastewater applied	
1,535,610.00	*gallons*
56.55	*acre-inches*
2.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alf 3 - 06/01/2023: Corn, silage

Field name: Alf 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.31	28.67	624.71	4,824.30
Fresh water	7.96	0.00	0.00	409.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.27	28.67	624.71	5,234.27
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	234.84	30.90	253.38	1,049.06
Nutrient balance	64.43	-2.23	371.33	4,185.22
Applied to removed ratio	1.27	0.93	2.47	4.99

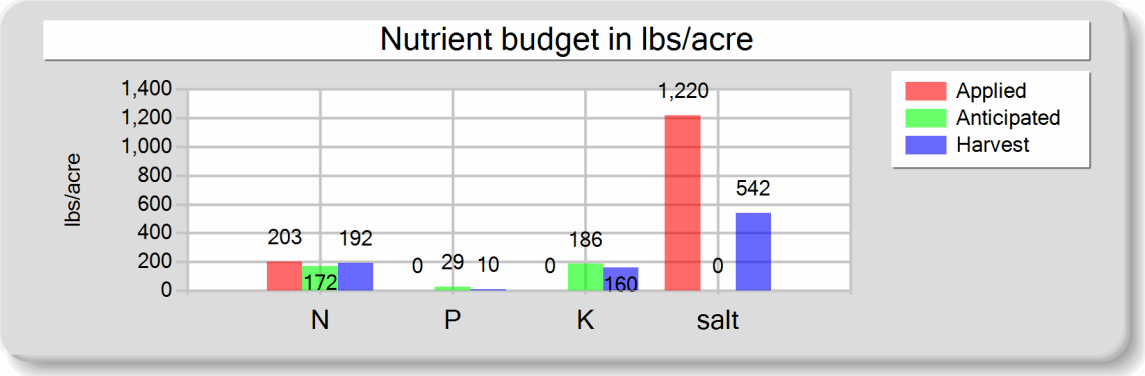
Fresh water applied	
17,950,487.00	*gallons*
661.06	*acre-inches*
34.79	*inches/acre*

Process wastewater applied	
1,508,485.00	*gallons*
55.55	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alm 06 - 01/01/2023: Almond, in shell

Field name: Alm 06 Crop: Almond, in shell Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	173.70	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	14.89	0.00	0.00	1,220.06
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	202.59	0.00	0.00	1,220.06
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	192.00	9.75	159.75	541.56
Nutrient balance	10.59	-9.75	-159.75	678.50
Applied to removed ratio	1.06	0.00	0.00	2.25

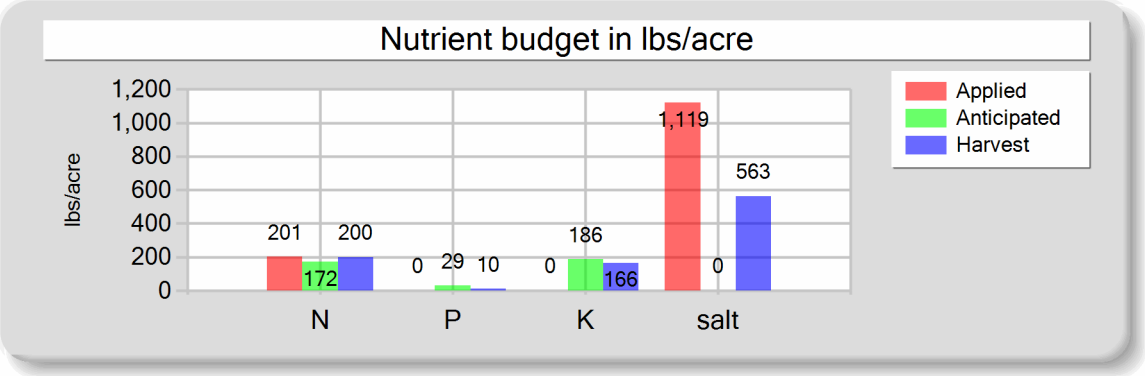
Fresh water applied	
120,994,920.00	*gallons*
4,455.83	*acre-inches*
37.13	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Alm 13 - 01/01/2023: Almond, in shell

Field name: Alm 13 Crop: Almond, in shell Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	173.70	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.44	0.00	0.00	1,118.88
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	201.14	0.00	0.00	1,118.88
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	199.68	10.14	166.14	563.22
Nutrient balance	1.46	-10.14	-166.14	555.66
Applied to removed ratio	1.01	0.00	0.00	1.99

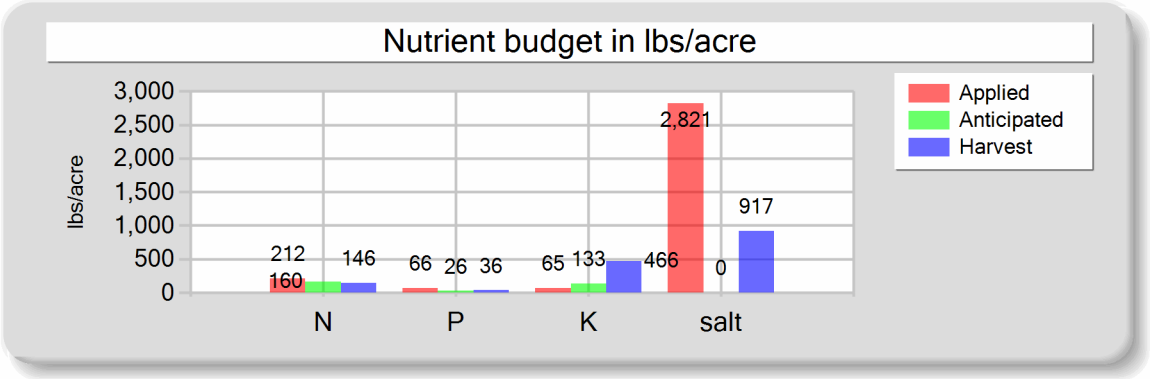
Fresh water applied	
114,110,903.00	*gallons*
4,202.32	*acre-inches*
37.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Delyo - 11/05/2022: Oats, silage-soft dough

Field name: Delyo | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	202.08	65.70	64.86	2,694.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.47	0.00	0.00	127.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.55	65.70	64.86	2,821.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	145.96	35.60	466.36	916.77
Nutrient balance	65.59	30.10	-401.50	1,904.68
Applied to removed ratio	1.45	1.85	0.14	3.08

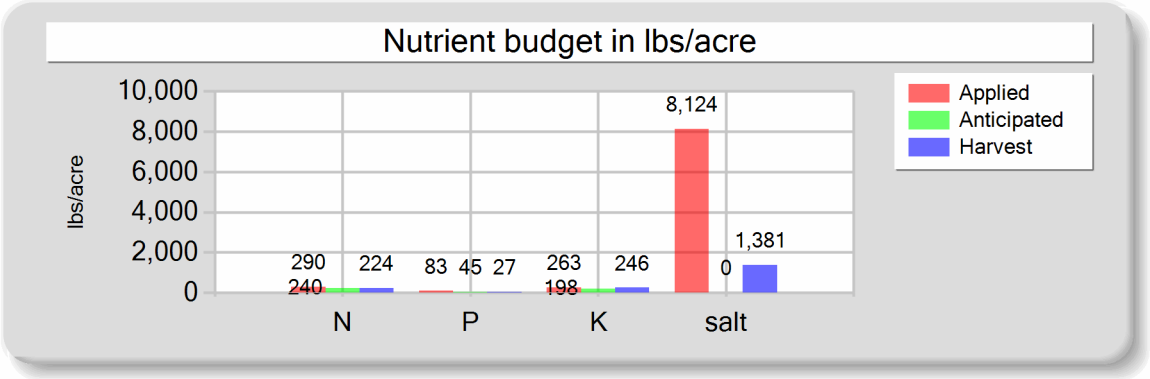
Fresh water applied	
5,270,670.00	*gallons*
194.10	*acre-inches*
10.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Delyo - 06/11/2023: Corn, silage

Field name: Delyo | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.64	82.58	263.30	7,707.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.09	0.00	0.00	416.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.73	82.58	263.30	8,123.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.28	26.70	245.64	1,380.67
Nutrient balance	65.45	55.88	17.66	6,743.20
Applied to removed ratio	1.29	3.09	1.07	5.88

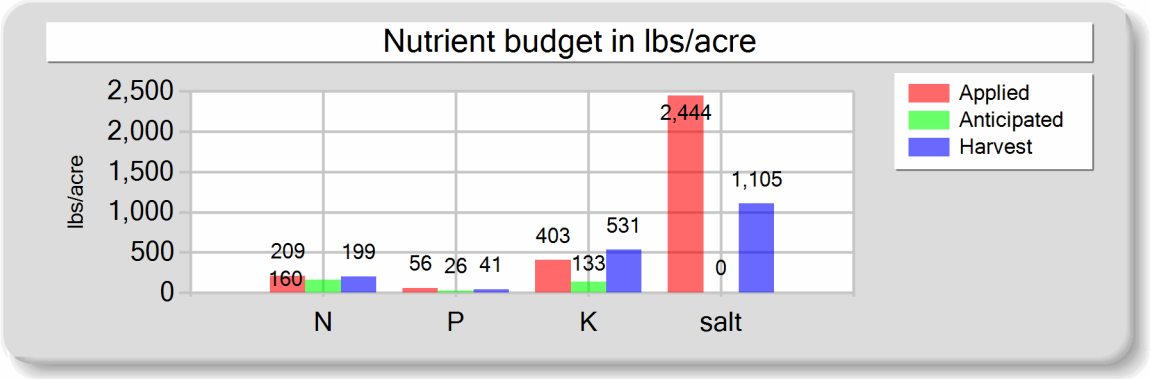
Fresh water applied	
17,275,554.00	*gallons*
636.20	*acre-inches*
35.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dry Cows - 10/26/2022: Oats, silage-soft dough

Field name: Dry Cows | Crop: Oats, silage-soft dough | Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.49	0.00	0.00	128.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.18	55.65	403.47	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.42	40.56	530.66	1,105.40
Nutrient balance	9.76	15.09	-127.19	1,338.70
Applied to removed ratio	1.05	1.37	0.76	2.21

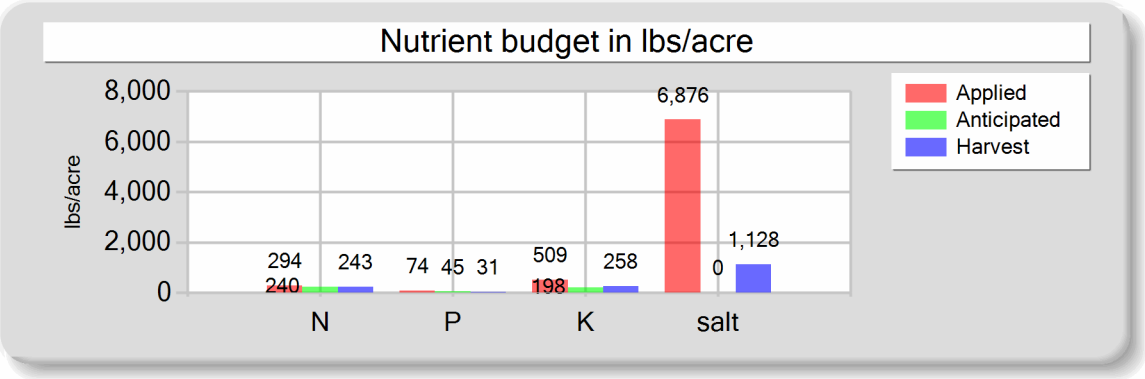
Fresh water applied	
9,145,837.00	*gallons*
336.81	*acre-inches*
10.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dry Cows - 06/02/2023: Corn, silage

Field name: Dry Cows | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.11	0.00	0.00	417.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.39	73.71	508.95	6,875.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	242.52	30.96	258.00	1,128.43
Nutrient balance	51.87	42.75	250.95	5,747.41
Applied to removed ratio	1.21	2.38	1.97	6.09

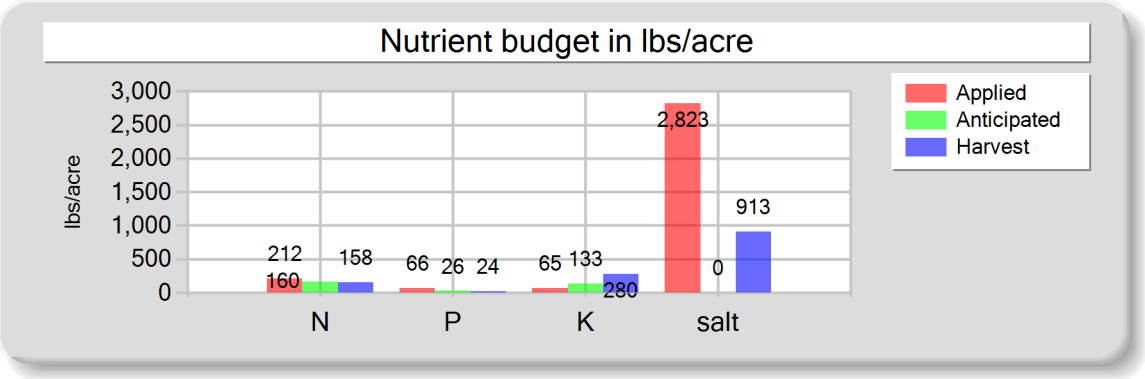
Fresh water applied	
29,821,628.00	*gallons*
1,098.23	*acre-inches*
35.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 11/03/2022: Oats, silage-soft dough

Field name: H1 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	202.08	65.70	64.86	2,694.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.50	0.00	0.00	128.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.58	65.70	64.86	2,823.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.08	24.32	279.68	913.36
Nutrient balance	53.50	41.38	-214.82	1,909.65
Applied to removed ratio	1.34	2.70	0.23	3.09

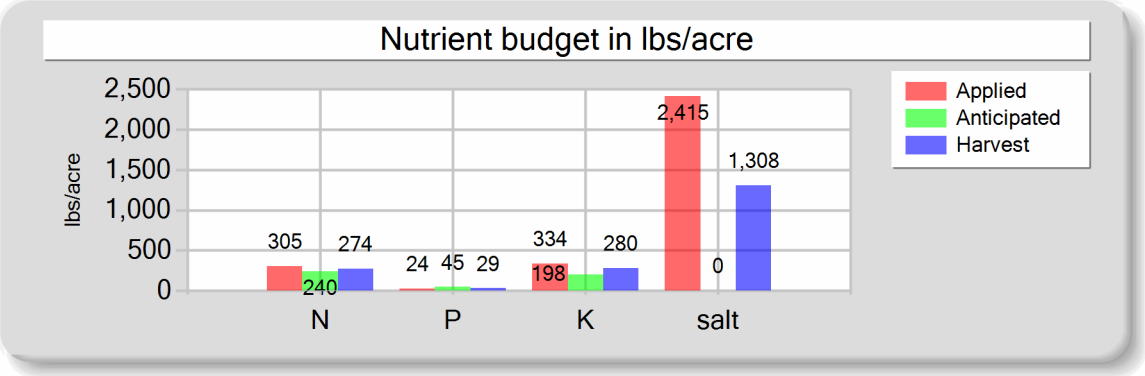
Fresh water applied	
8,892,690.00	*gallons*
327.49	*acre-inches*
10.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 06/09/2023: Corn, silage

Field name: H1 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.86	23.68	334.12	2,019.27
Fresh water	7.69	0.00	0.00	396.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.55	23.68	334.12	2,415.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.48	29.20	280.32	1,307.72
Nutrient balance	30.07	-5.52	53.80	1,107.72
Applied to removed ratio	1.11	0.81	1.19	1.85

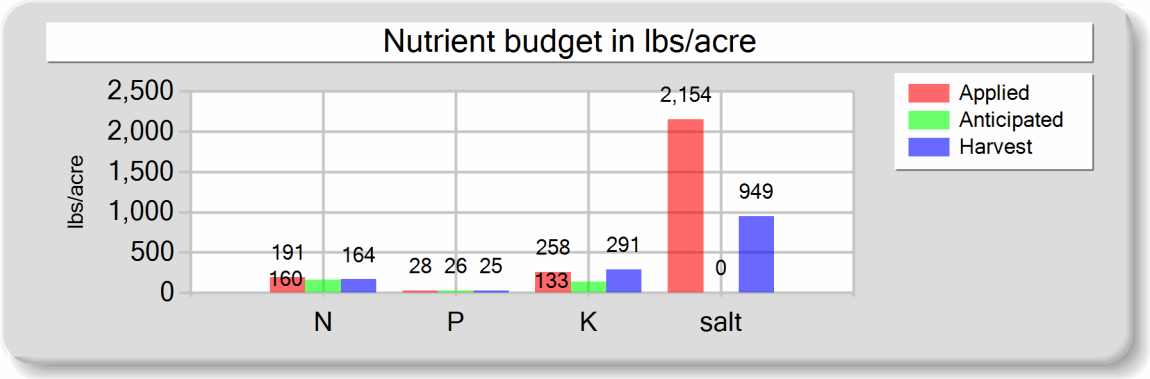
Fresh water applied	
27,388,640.00	*gallons*
1,008.63	*acre-inches*
33.62	*inches/acre*

Process wastewater applied	
1,255,255.00	*gallons*
46.23	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

H2 - 11/04/2022: Oats, silage-soft dough

Field name: H2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.78	27.74	258.18	2,024.24
Fresh water	2.51	0.00	0.00	129.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.29	27.74	258.18	2,153.61
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	164.32	25.28	290.72	949.42
Nutrient balance	26.97	2.46	-32.54	1,204.20
Applied to removed ratio	1.16	1.10	0.89	2.27

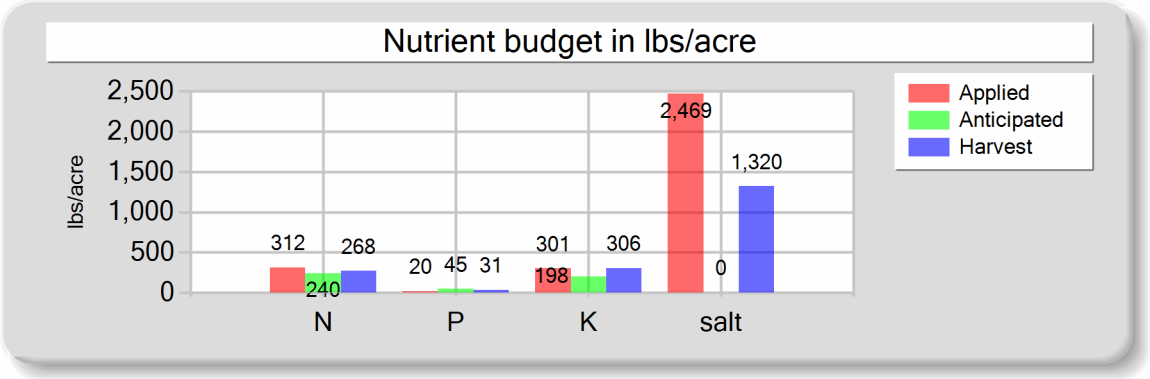
Fresh water applied	
15,205,038.00	*gallons*
559.95	*acre-inches*
10.98	*inches/acre*

Process wastewater applied	
2,662,936.00	*gallons*
98.07	*acre-inches*
1.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

H2 - 06/09/2023: Corn, silage

Field name: H2 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	233.54	20.10	300.71	2,075.89
Fresh water	7.64	0.00	0.00	393.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.18	20.10	300.71	2,469.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.32	31.20	305.76	1,320.13
Nutrient balance	43.86	-11.10	-5.05	1,149.12
Applied to removed ratio	1.16	0.64	0.98	1.87

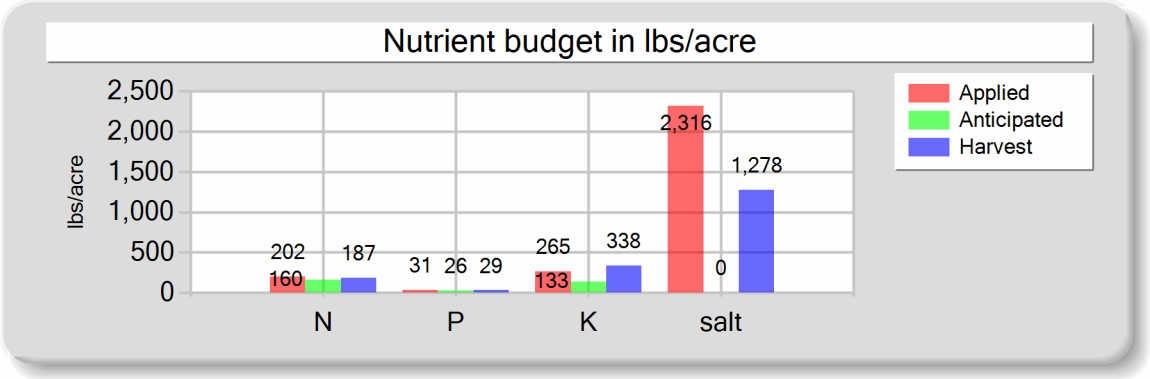
Fresh water applied	
46,231,602.00	*gallons*
1,702.55	*acre-inches*
33.38	*inches/acre*

Process wastewater applied	
1,482,088.00	*gallons*
54.58	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

H3 - 11/03/2022: Oats, silage-soft dough

Field name: H3 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.16	30.80	264.68	2,183.78
Fresh water	2.56	0.00	0.00	131.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.72	30.80	264.68	2,315.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	186.76	28.98	338.10	1,277.57
Nutrient balance	14.96	1.82	-73.42	1,037.95
Applied to removed ratio	1.08	1.06	0.78	1.81

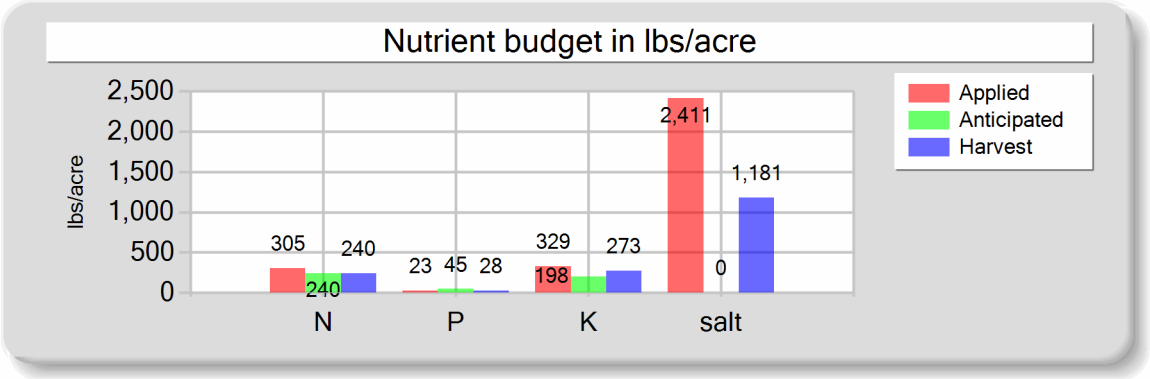
Fresh water applied	
14,876,204.00	*gallons*
547.84	*acre-inches*
11.18	*inches/acre*

Process wastewater applied	
2,614,273.00	*gallons*
96.27	*acre-inches*
1.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

H3 - 06/10/2023: Corn, silage

Field name: H3 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.94	23.15	328.71	2,018.55
Fresh water	7.63	0.00	0.00	392.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.57	23.15	328.71	2,411.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.94	27.90	273.42	1,180.50
Nutrient balance	64.63	-4.75	55.29	1,230.93
Applied to removed ratio	1.27	0.83	1.20	2.04

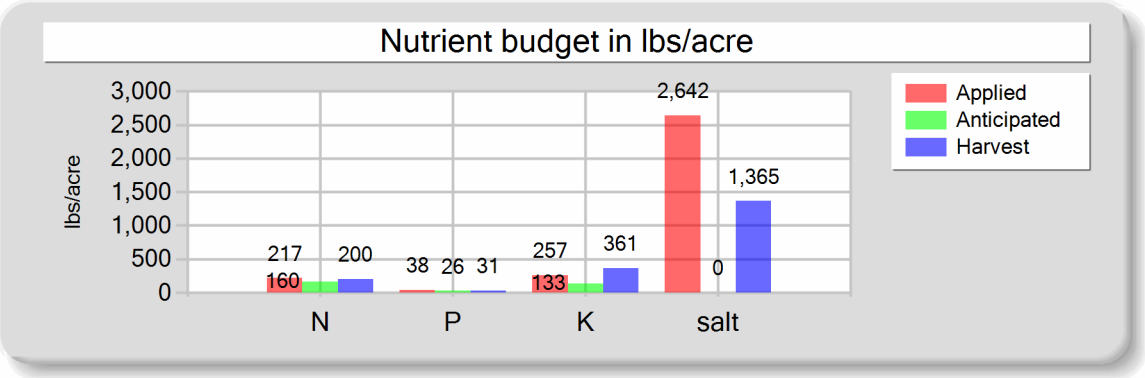
Fresh water applied	
44,364,698.00	*gallons*
1,633.80	*acre-inches*
33.34	*inches/acre*

Process wastewater applied	
1,964,931.00	*gallons*
72.36	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

H4 - 11/04/2022: Oats, silage-soft dough

Field name: H4 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.55	38.48	256.56	2,514.46
Fresh water	2.48	0.00	0.00	127.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.03	38.48	256.56	2,642.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.52	30.96	361.20	1,364.85
Nutrient balance	17.51	7.52	-104.64	1,277.45
Applied to removed ratio	1.09	1.24	0.71	1.94

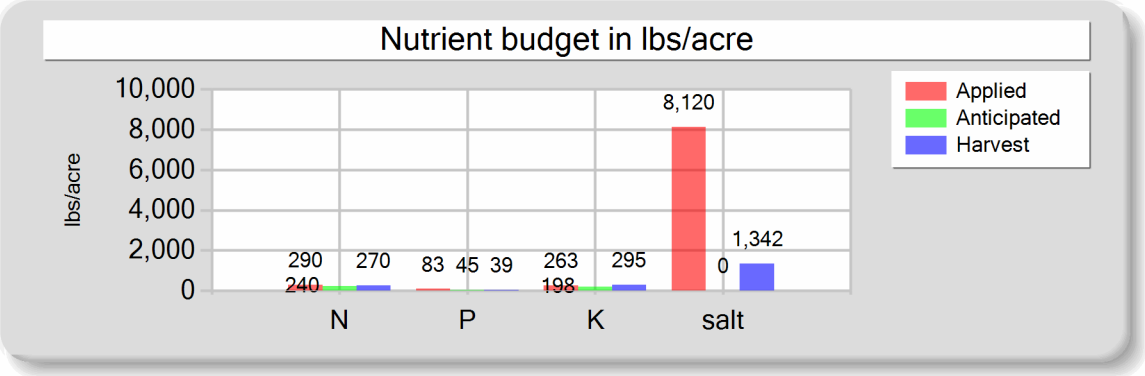
Fresh water applied	
3,829,943.00	*gallons*
141.04	*acre-inches*
10.85	*inches/acre*

Process wastewater applied	
663,910.00	*gallons*
24.45	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

H4 - 06/11/2023: Corn, silage

Field name: H4 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.64	82.58	263.30	7,707.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.01	0.00	0.00	412.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.65	82.58	263.30	8,119.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.64	38.52	295.32	1,341.60
Nutrient balance	20.01	44.06	-32.02	6,778.25
Applied to removed ratio	1.07	2.14	0.89	6.05

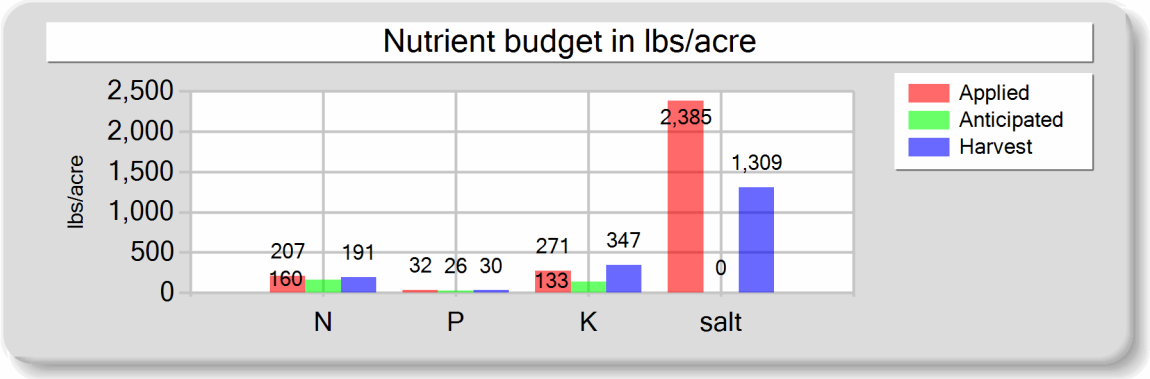
Fresh water applied	
12,356,331.00	*gallons*
455.04	*acre-inches*
35.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H5 - 11/04/2022: Oats, silage-soft dough

Field name: H5 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.82	32.00	270.82	2,256.94
Fresh water	2.49	0.00	0.00	128.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.32	32.00	270.82	2,385.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	191.40	29.70	346.50	1,309.31
Nutrient balance	15.92	2.30	-75.68	1,075.98
Applied to removed ratio	1.08	1.08	0.78	1.82

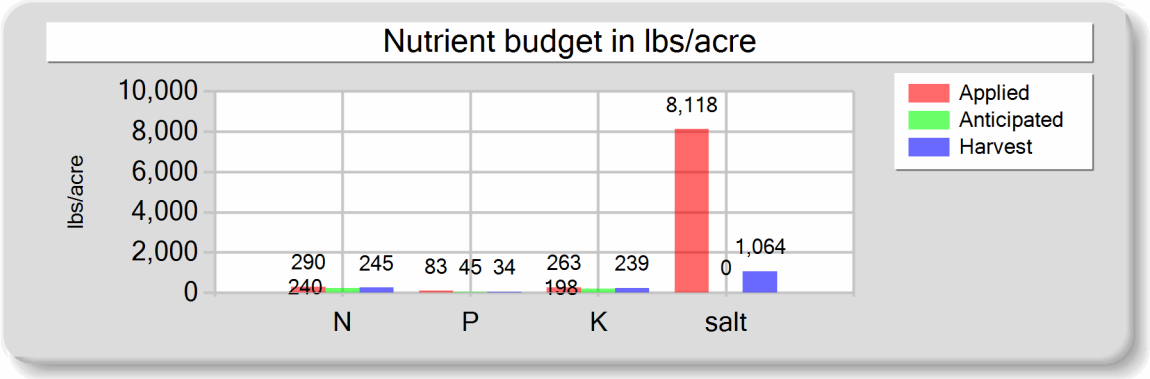
Fresh water applied	
4,732,320.00	*gallons*
174.28	*acre-inches*
10.89	*inches/acre*

Process wastewater applied	
872,835.00	*gallons*
32.14	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

H5 - 06/10/2023: Corn, silage

Field name: H5 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.64	82.58	263.30	7,707.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.97	0.00	0.00	410.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.60	82.58	263.30	8,117.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.10	34.20	239.40	1,063.96
Nutrient balance	44.50	48.38	23.90	7,053.65
Applied to removed ratio	1.18	2.41	1.10	7.63

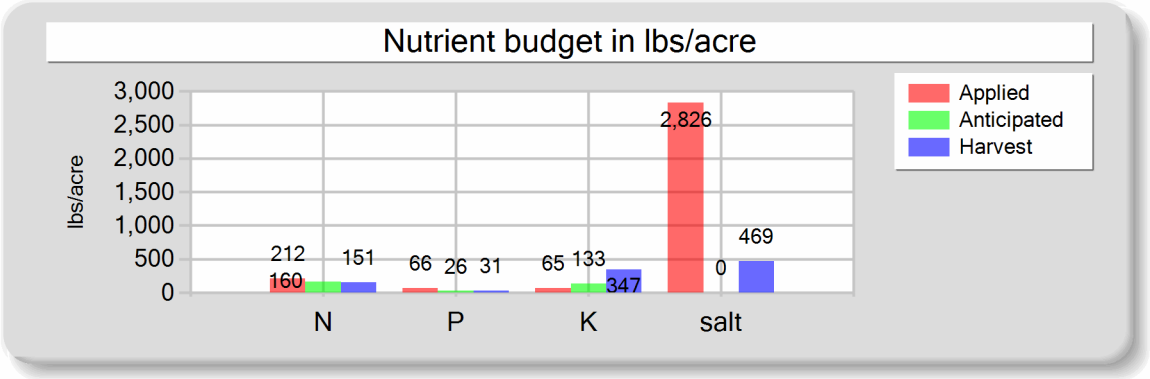
Fresh water applied	
15,125,440.00	*gallons*
557.02	*acre-inches*
34.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H6 - 11/05/2022: Oats, silage-soft dough

Field name: H6 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	202.08	65.70	64.86	2,694.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.56	0.00	0.00	131.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.63	65.70	64.86	2,825.94
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	151.36	30.96	347.44	469.50
Nutrient balance	60.27	34.74	-282.58	2,356.44
Applied to removed ratio	1.40	2.12	0.19	6.02

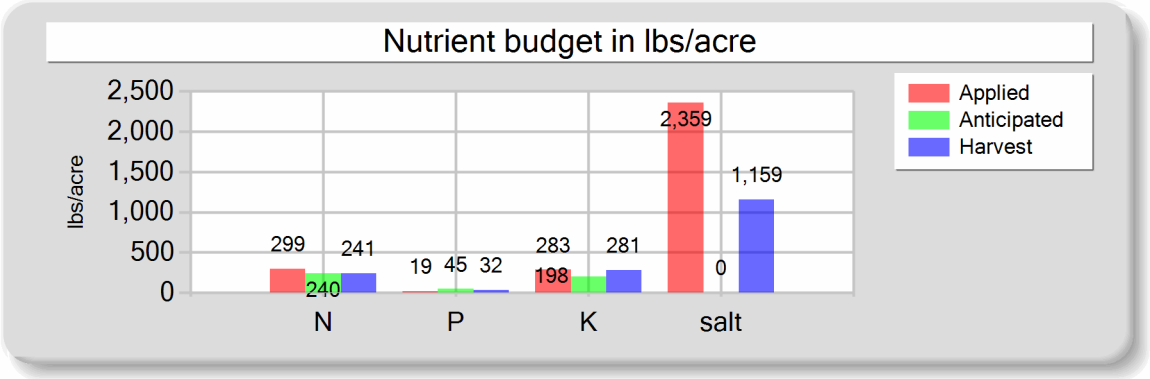
Fresh water applied
6,063,120.00 *gallons*
223.28 *acre-inches*
11.16 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

H6 - 06/11/2023: Corn, silage

Field name: H6 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.17	18.95	283.50	1,957.04
Fresh water	7.81	0.00	0.00	402.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.98	18.95	283.50	2,359.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.08	31.57	281.26	1,159.25
Nutrient balance	57.90	-12.62	2.24	1,199.91
Applied to removed ratio	1.24	0.60	1.01	2.04

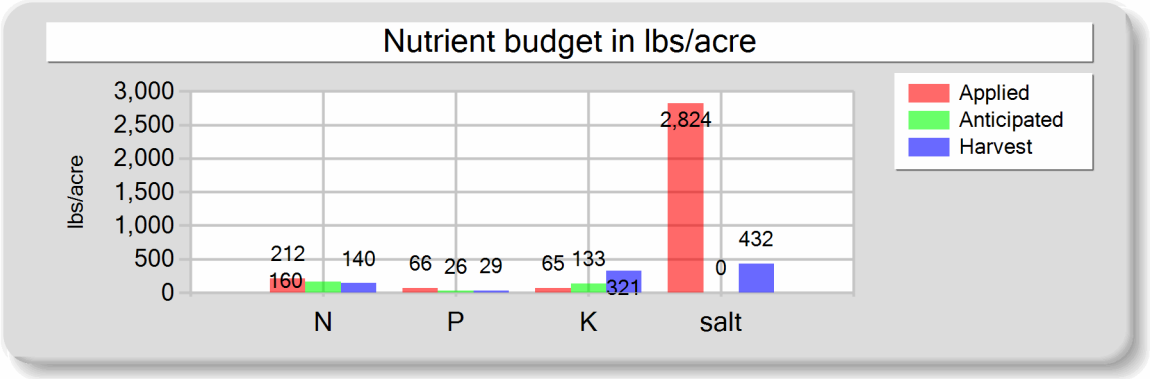
Fresh water applied	
18,533,560.00	*gallons*
682.53	*acre-inches*
34.13	*inches/acre*

Process wastewater applied	
547,935.00	*gallons*
20.18	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

H7 - 11/05/2022: Oats, silage-soft dough

Field name: H7 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	202.08	65.70	64.86	2,694.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.52	0.00	0.00	129.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.60	65.70	64.86	2,823.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.92	28.62	321.18	432.37
Nutrient balance	71.68	37.08	-256.32	2,391.58
Applied to removed ratio	1.51	2.30	0.20	6.53

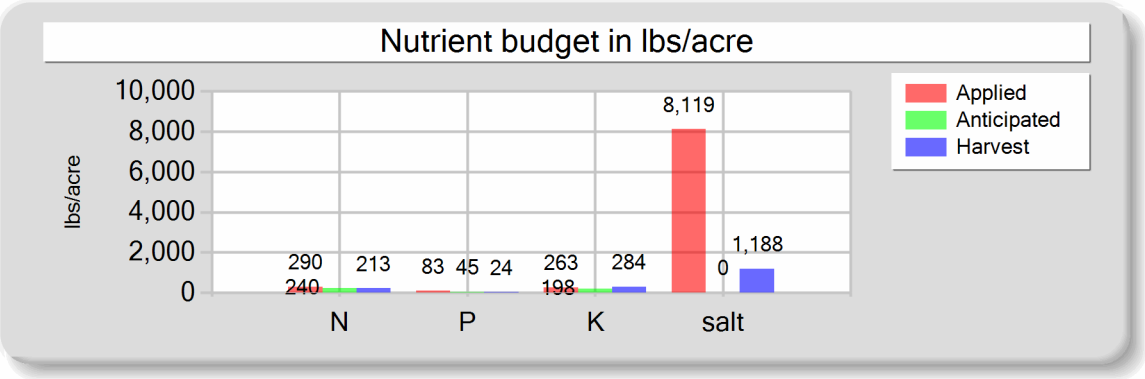
Fresh water applied	
10,151,584.00	*gallons*
373.85	*acre-inches*
11.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H7 - 06/11/2023: Corn, silage

Field name: H7 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.64	82.58	263.30	7,707.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.99	0.00	0.00	411.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.63	82.58	263.30	8,118.76
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	213.12	23.68	284.16	1,187.59
Nutrient balance	76.51	58.90	-20.86	6,931.18
Applied to removed ratio	1.36	3.49	0.93	6.84

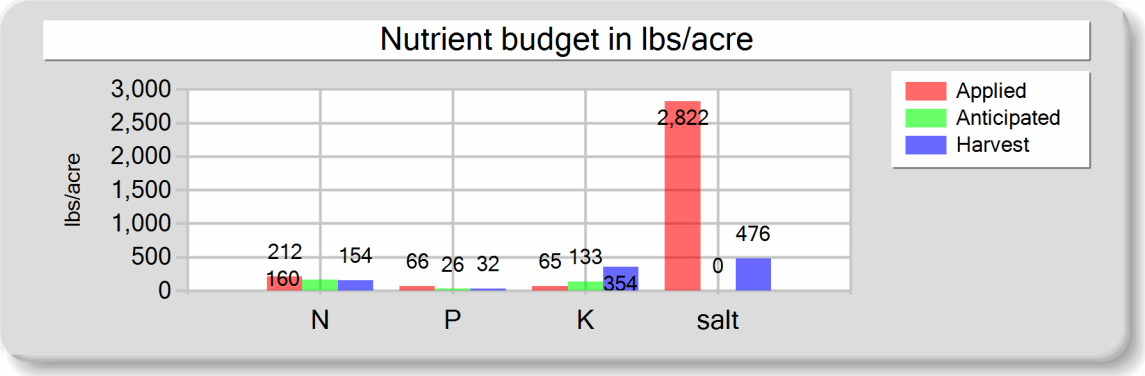
Fresh water applied	
32,231,762.00	*gallons*
1,186.99	*acre-inches*
34.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H8 - 11/05/2022: Oats, silage-soft dough

Field name: H8 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	202.08	65.70	64.86	2,694.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.47	0.00	0.00	127.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.55	65.70	64.86	2,821.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.00	31.50	353.50	475.88
Nutrient balance	57.55	34.20	-288.64	2,345.89
Applied to removed ratio	1.37	2.09	0.18	5.93

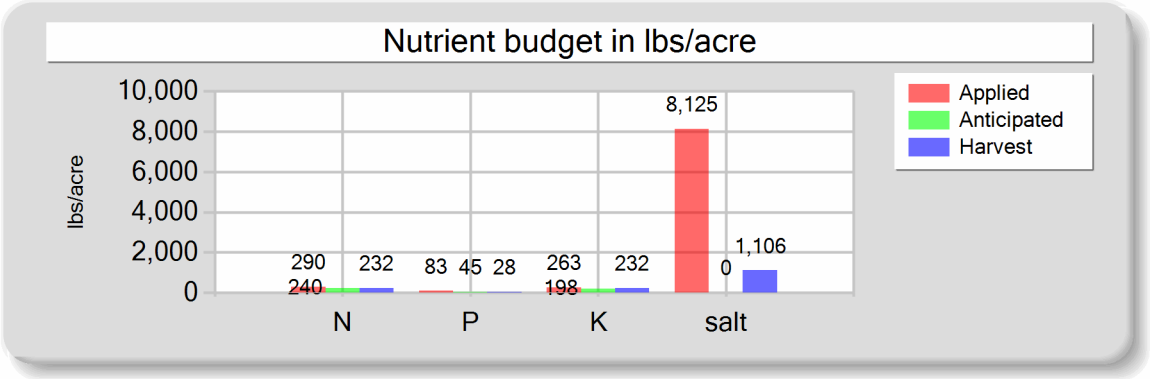
Fresh water applied	
3,229,028.00	*gallons*
118.91	*acre-inches*
10.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H8 - 06/11/2023: Corn, silage

Field name: H8 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	210.64	82.58	263.30	7,707.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.10	0.00	0.00	417.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.74	82.58	263.30	8,124.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.84	27.60	231.84	1,106.47
Nutrient balance	57.90	54.98	31.46	7,018.19
Applied to removed ratio	1.25	2.99	1.14	7.34

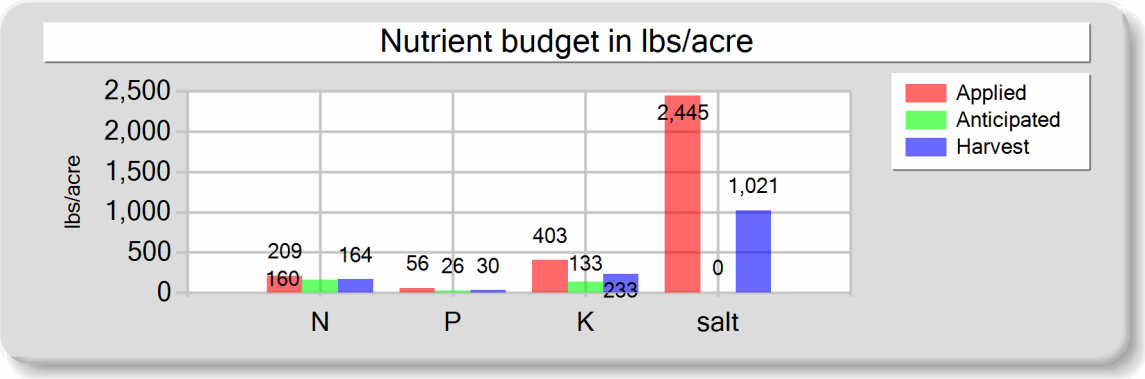
Fresh water applied	
10,577,556.00	*gallons*
389.54	*acre-inches*
35.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pasture - 10/27/2022: Oats, silage-soft dough

Field name: Pasture | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.50	0.00	0.00	128.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.19	55.65	403.47	2,444.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	164.00	29.52	232.88	1,021.30
Nutrient balance	45.19	26.13	170.59	1,423.46
Applied to removed ratio	1.28	1.89	1.73	2.39

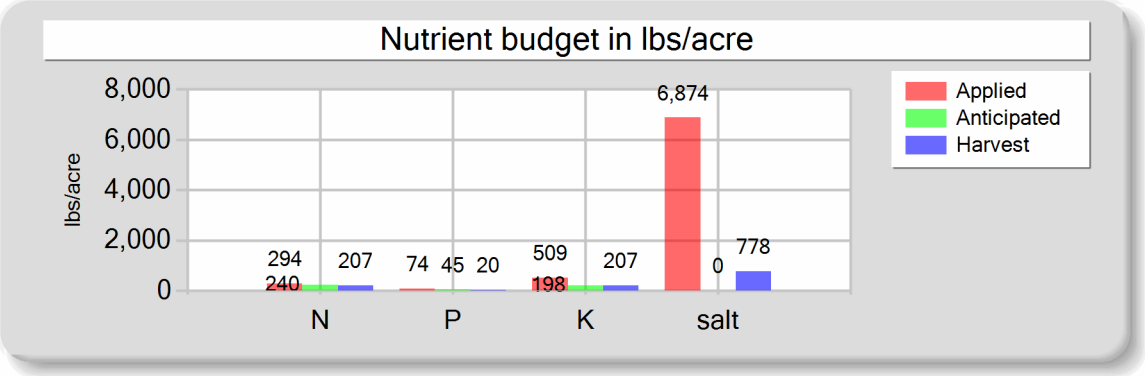
Fresh water applied	
4,744,736.00	*gallons*
174.73	*acre-inches*
10.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pasture - 06/02/2023: Corn, silage

Field name: Pasture | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.08	0.00	0.00	416.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.36	73.71	508.95	6,874.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	206.64	19.68	206.64	778.31
Nutrient balance	87.72	54.03	302.31	6,096.09
Applied to removed ratio	1.42	3.75	2.46	8.83

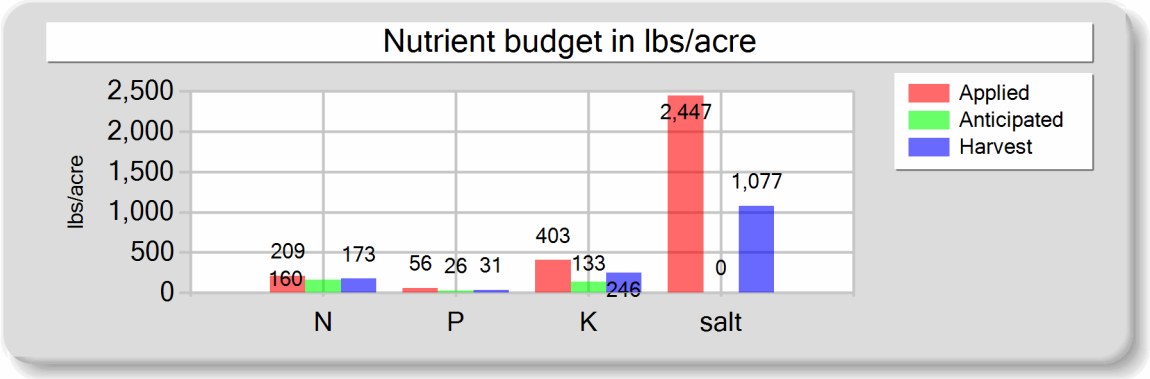
Fresh water applied	
15,338,704.00	*gallons*
564.87	*acre-inches*
35.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The Entrance - 10/27/2022: Oats, silage-soft dough

Field name: The Entrance | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.69	55.65	403.47	2,316.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.55	0.00	0.00	131.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.24	55.65	403.47	2,447.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.00	31.14	245.66	1,077.35
Nutrient balance	36.24	24.51	157.81	1,369.99
Applied to removed ratio	1.21	1.79	1.64	2.27

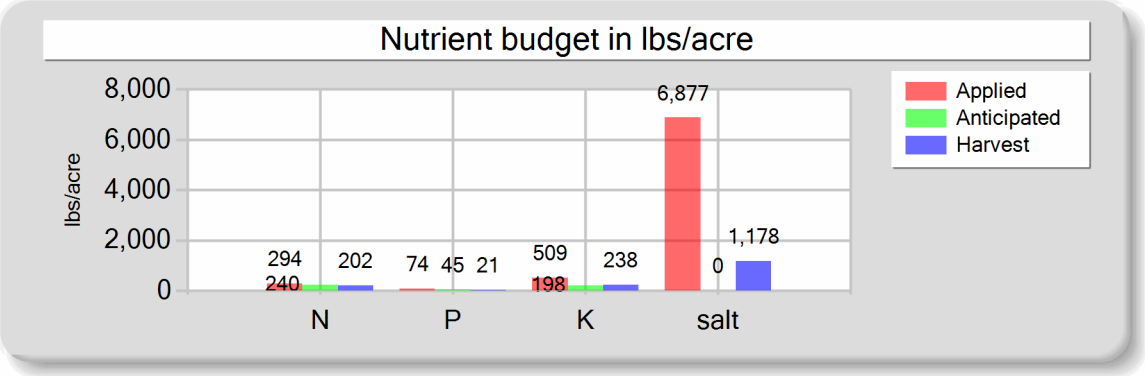
Fresh water applied	
9,982,368.00	*gallons*
367.62	*acre-inches*
11.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The Entrance - 06/02/2023: Corn, silage

Field name: The Entrance | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	215.28	73.71	508.95	6,458.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.13	0.00	0.00	418.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.41	73.71	508.95	6,877.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.02	20.72	238.28	1,177.85
Nutrient balance	92.39	52.99	270.67	5,699.20
Applied to removed ratio	1.46	3.56	2.14	5.84

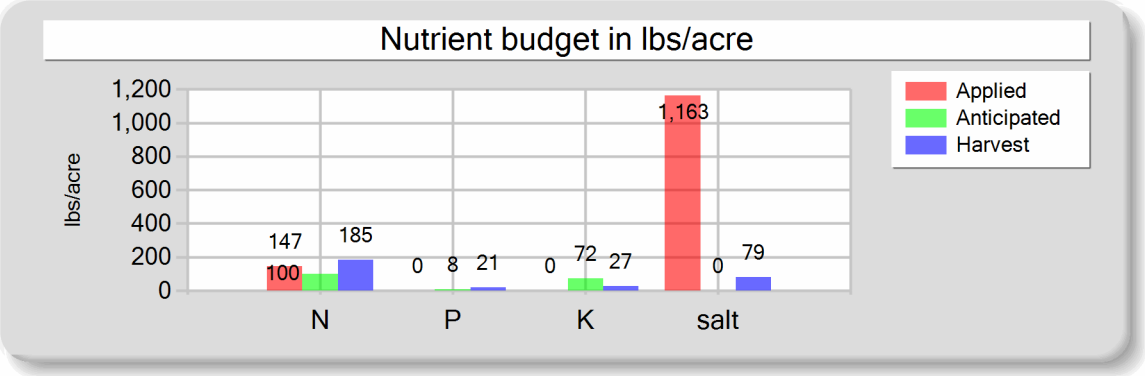
Fresh water applied	
31,837,377.00	*gallons*
1,172.46	*acre-inches*
35.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Wal 07 - 01/01/2023: Walnuts

Field name: Wal 07 Crop: Walnuts Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	118.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	14.46	0.00	0.00	1,163.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	146.86	0.00	0.00	1,163.18
Anticipated crop nutrient removal	100.00	7.50	72.00	0.00
Actual crop nutrient removal	185.28	21.12	27.36	79.46
Nutrient balance	-38.42	-21.12	-27.36	1,083.72
Applied to removed ratio	0.79	0.00	0.00	14.64

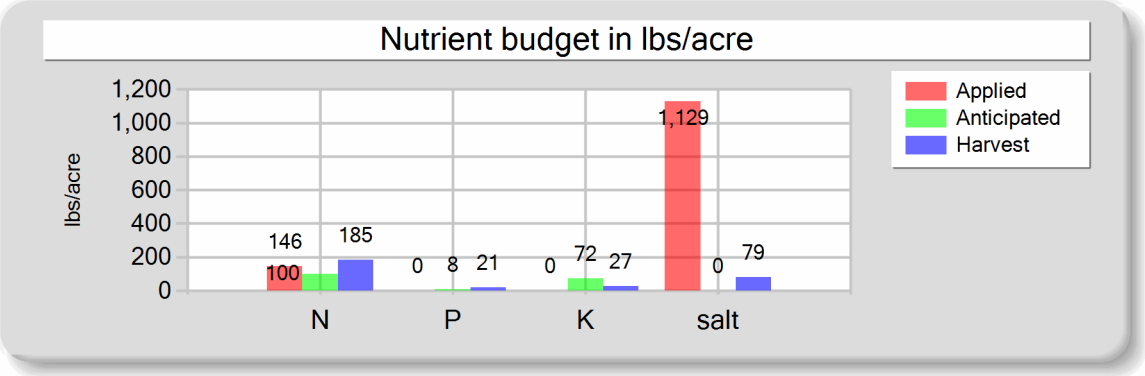
Fresh water applied	
49,780,800.00	*gallons*
1,833.26	*acre-inches*
36.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Wal 15 - 01/01/2023: Walnuts

Field name: Wal 15 Crop: Walnuts Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	118.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	14.03	0.00	0.00	1,128.86
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	146.43	0.00	0.00	1,128.86
Anticipated crop nutrient removal	100.00	7.50	72.00	0.00
Actual crop nutrient removal	185.28	21.12	27.36	79.46
Nutrient balance	-38.85	-21.12	-27.36	1,049.40
Applied to removed ratio	0.79	0.00	0.00	14.21

Fresh water applied	
84,063,228.00	*gallons*
3,095.76	*acre-inches*
35.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 MLC Heifer Corral Solids

Sample and source description: 2022 MLC Heifer Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	6,800.00	49,300.00							28.30
DL	100.00	100.00	100.00							0.01

2022 Fall Corral Solids

Sample and source description: 2022 Fall Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	11,500.00	40,500.00							36.00
DL	100.00	100.00	100.00							0.01

2022 Fall Settling Basin

Sample and source description: 2022 Fall Settling Basin

Sample date: 09/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	7,900.00	7,800.00							32.40
DL	100.00	100.00	100.00							0.01

MLC Heifer Spring Corral Solids

Sample and source description: MLC Heifer Spring Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	6,300.00	43,500.00	15,100.00	6,700.00	2,300.00	4,100.00	2,620.00		55.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	6,900.00	22,000.00	29,700.00	11,400.00	5,400.00	4,400.00	2,000.00		64.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

MLC Heifer Fall Corral Solids

Sample and source description: MLC Heifer Fall Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	5,400.00	36,100.00							52.00
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	7,000.00	26,600.00							44.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

2022 MLC WWSQ4

Sample and source description: 2022 MLC WWSQ4

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	307.00	147.00	0.00	0.00	51.10	1,450.00								7,880.00	5,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 10/26/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	369.00	244.00	0.00	0.00	45.30	586.00								5,310.00	3,780
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

MLC WWSQ1

Sample and source description: MLC WWSQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	302.00	176.00	0.00	0.00	33.90	546.00	110.00	84.20	163.00	2,580.00	0.00	176.00	476.00	6,280.00	7,920
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	487.00	339.00	0.00	0.00	90.30	602.00	205.00	122.00	183.00	2,880.00	0.00	148.00	199.00	5,920.00	5,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

MLC WWSQ2

Sample and source description: MLC WWSQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	311.00	176.00	0.00	0.00	40.80	990.00								6,090.00	7,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	437.00	316.00	0.00	0.00	57.80	769.00								5,600.00	3,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	963.00	599.00	0.00	0.00	82.90	1,240.00								8,860.00	8,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

MLC WWSQ4

Sample and source description: MLC WWSQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	454.00	319.00	0.00	0.00	57.20	677.00								5,960.00	4,580
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	395.00	353.00	0.00	0.00	42.80	605.00								6,030.00	6,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

OID Canal

Sample description: OID Canal

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	1.01								70.90	52
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

IW #1

Sample description: IW #1

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.69	0.00	1.69	16.90	9.65	15.00	92.00	0.00	8.17	3.63	232.00	140
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

IW Alm 13

Sample description: IW Alm 13

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.72	0.00	1.72	16.90	9.58	14.90	96.00	0.00	8.37	4.03	233.00	150
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

Irrigation Well

IW Alm 6

Sample description: IW Alm 6

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.77	0.00	1.77	17.10	9.71	15.20	94.00	0.00	7.41	2.37	233.00	145
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

IW Walnut

Sample description: IW Walnut

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.74	0.00	1.74	17.10	9.65	15.10	96.00	0.00	8.28	3.24	231.00	140
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

"17" - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,100.00	15,300.00		14.40
DL	100.00	100.00	100.00		0.01

"17" - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	500.00	4,500.00		4.07
DL	100.00	100.00	100.00		0.01

"25" - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,100.00	15,300.00		14.40
DL	100.00	100.00	100.00		0.01

"25" - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	500.00	4,200.00		4.82
DL	100.00	100.00	100.00		0.01

Alf 1 - 10/25/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,100.00	14,800.00		12.90
DL	100.00	100.00	100.00		0.01

Alf 1 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	500.00	4,000.00		5.51
DL	100.00	100.00	100.00		0.01

Alf 2 - 10/25/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,100.00	14,800.00		12.90
DL	100.00	100.00	100.00		0.01

Alf 2 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	500.00	4,100.00		4.85
DL	100.00	100.00	100.00		0.01

Alf 3 - 10/25/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,100.00	14,800.00		12.90
DL	100.00	100.00	100.00		0.01

Alf 3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	600.00	4,400.00		4.36
DL	100.00	100.00	100.00		0.01

Delyo - 11/05/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	13,100.00		11.60
DL	100.00	100.00	100.00		0.01

Delyo - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	500.00	4,600.00		6.84
DL	100.00	100.00	100.00		0.01

Dry Cows - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,200.00	15,700.00		14.60
DL	100.00	100.00	100.00		0.01

Dry Cows - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	600.00	5,000.00		5.44
DL	100.00	100.00	100.00		0.01

H1 - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	9,200.00		9.16
DL	100.00	100.00	100.00		0.01

H1 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	500.00	4,800.00		6.29
DL	100.00	100.00	100.00		0.01

H2 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	9,200.00		9.16
DL	100.00	100.00	100.00		0.01

H2 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	500.00	4,900.00		6.15
DL	100.00	100.00	100.00		0.01

H3 - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	900.00	10,500.00		10.90
DL	100.00	100.00	100.00		0.01

H3 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	400.00	4,100.00		5.55
DL	100.00	100.00	100.00		0.01

H4 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	900.00	10,500.00		10.90
DL	100.00	100.00	100.00		0.01

H4 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	600.00	4,600.00		5.87
DL	100.00	100.00	100.00		0.01

H5 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	900.00	10,500.00		10.90
DL	100.00	100.00	100.00		0.01

H5 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	4,200.00		5.10
DL	100.00	100.00	100.00		0.01

H6 - 11/05/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	900.00	10,100.00		5.27
DL	100.00	100.00	100.00		0.01

H6 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	550.00	4,900.00		5.61
DL	100.00	100.00	100.00		0.01

H7 - 11/05/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	900.00	10,100.00		5.27
DL	100.00	100.00	100.00		0.01

H7 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	400.00	4,800.00		6.23
DL	100.00	100.00	100.00		0.01

H8 - 11/05/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	900.00	10,100.00		5.27
DL	100.00	100.00	100.00		0.01

H8 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	500.00	4,200.00		5.76
DL	100.00	100.00	100.00		0.01

Pasture - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	900.00	7,100.00		9.64
DL	100.00	100.00	100.00		0.01

Pasture - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	400.00	4,200.00		4.37
DL	100.00	100.00	100.00		0.01

The Entrance - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	900.00	7,100.00		9.64
DL	100.00	100.00	100.00		0.01

The Entrance - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	400.00	4,600.00		6.61
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

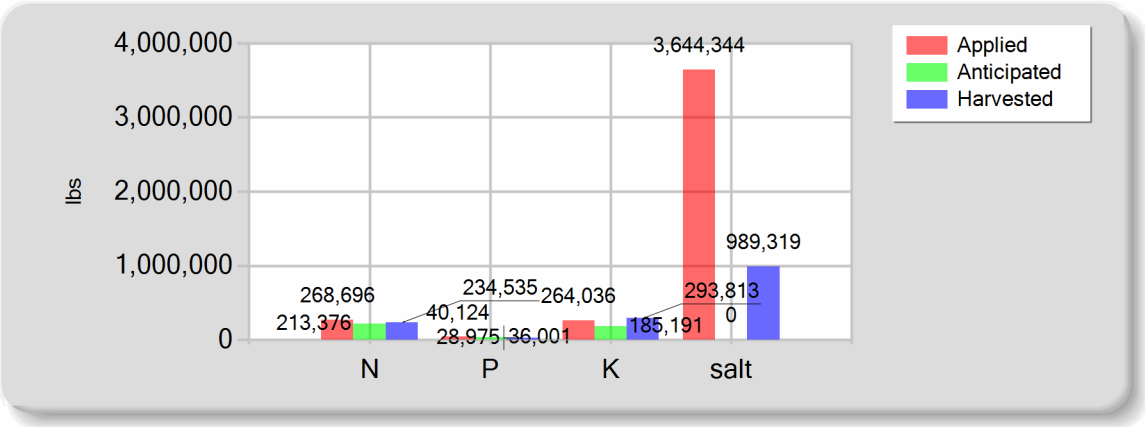
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

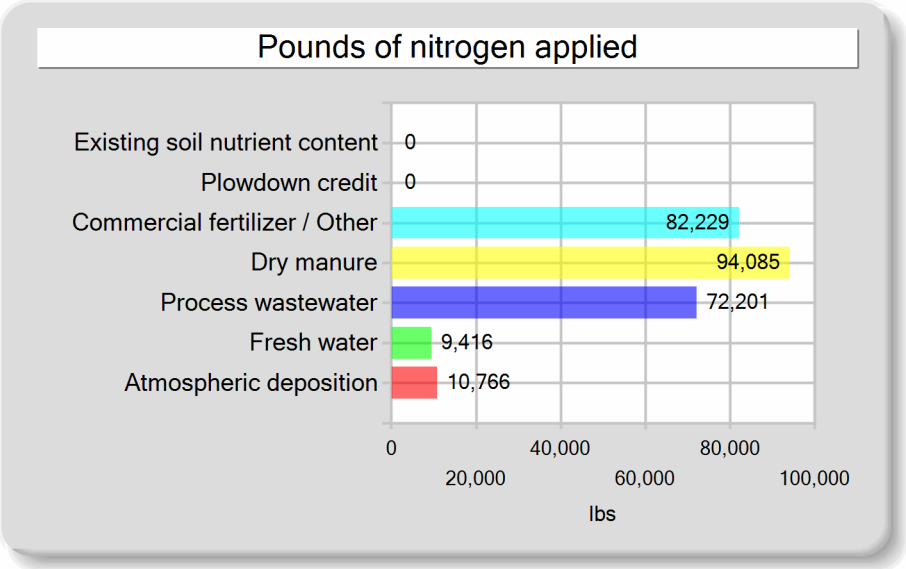
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

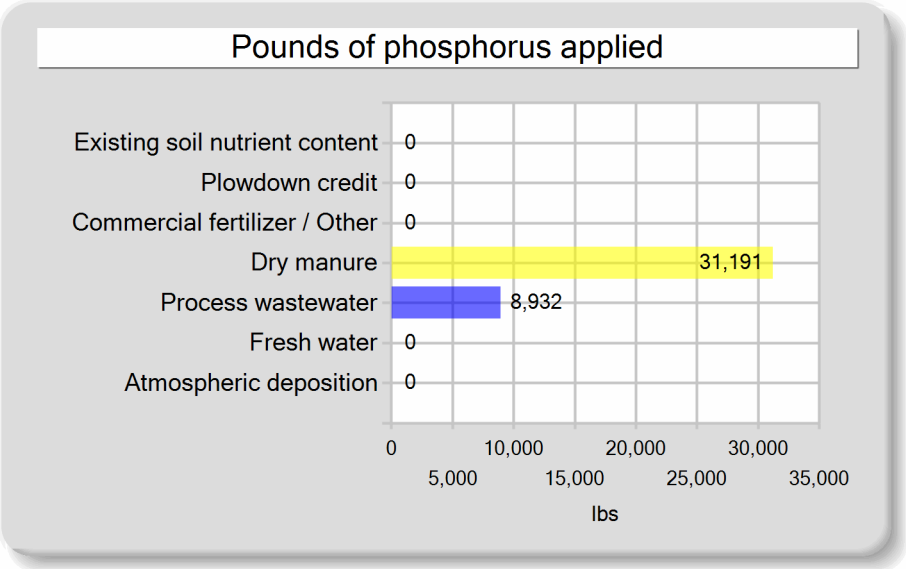
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	82,228.90	0.00	0.00	0.00
Dry manure	94,084.69	31,191.13	145,605.60	2,110,354.46
Process wastewater	72,200.64	8,932.50	118,429.94	890,288.13
Fresh water	9,415.72	0.00	0.00	643,701.28
Atmospheric deposition	10,766.00	0.00	0.00	0.00
Total nutrients applied	268,695.95	40,123.63	264,035.54	3,644,343.86
Anticipated crop nutrient removal	213,376.00	36,000.50	185,191.20	0.00
Actual crop nutrient removal	234,535.40	28,975.43	293,812.92	989,319.02
Nutrient balance	34,160.55	11,148.20	-29,777.38	2,655,024.84
Applied to removed ratio	1.15	1.38	0.90	3.68

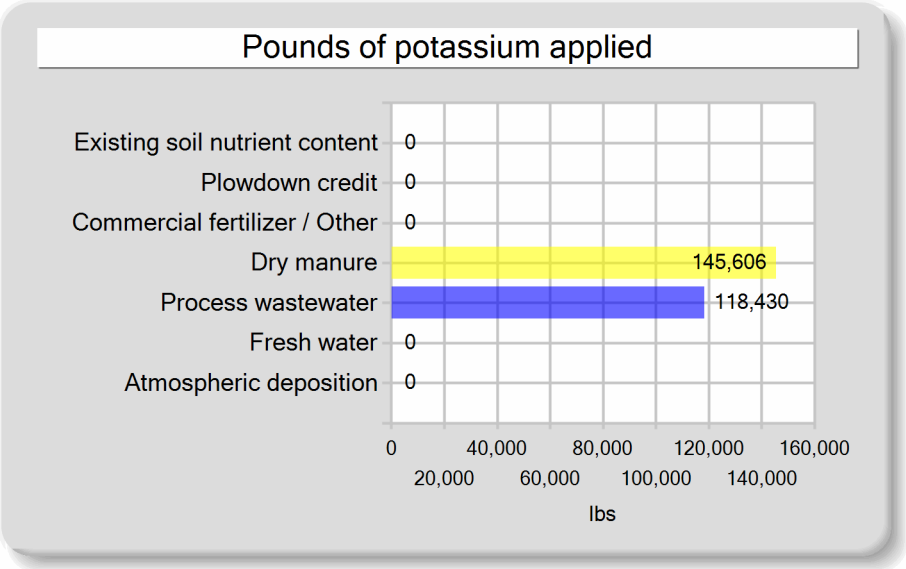
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

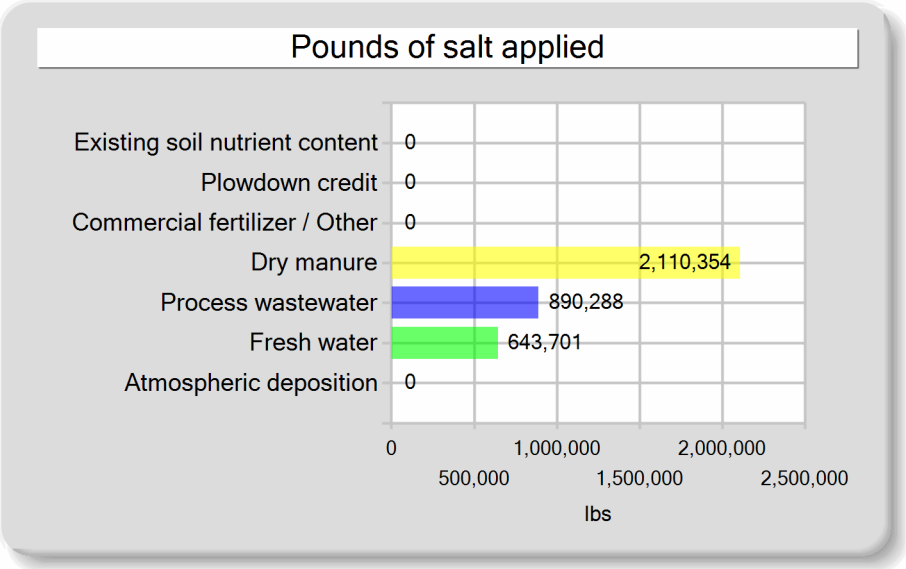


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Fields H7 lower than anticipated nitrogen removal for winter forage (8.93 lbs of nitrogen per acre removed vs 10 lbs of nitrogen per acre anticipated) resulted in these fields exceeding the 1.4 requirement.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

joe martin joe martin (Jun 25, 2024 07:27 PDT)	joe martin joe martin (Jun 25, 2024 07:27 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe B Martin	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	Jun 25, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW A	T3310046A	Electrical Conductivity (EC)	EPA 120.1 and	0.233	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.67	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	149	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)	3.82
Electrical Conductivity (EC)	99.5			0.431
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)	1.17

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 20.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW B	T3310046B	Electrical Conductivity (EC)	EPA 120.1 and	0.232	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.74	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	149	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)	3.82
Electrical Conductivity (EC)	99.5			0.431
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)	1.17

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation /Ground Water (H14)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 21.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 1	T3310046C	Electrical Conductivity (EC)	EPA 120.1 and	0.232	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	92.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	8.17	mg/L
		Sodium (Na)	EPA 200.7	15.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	9.65	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.69	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	140	mg/L
		Calcium (Ca)	EPA 200.7	16.9	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	3.63	mg/L
		Soluble Salts (SALT-SOL)	Calculation	149	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)			3.82
Calcium (Ca)	92.1	99.5 (T3310046C)	99.9 (T3310046C)			0.329
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	101	94.4 (T3310046C)	95.2 (T3310046C)			0.810
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	92.0	95.8 (T3310046C)	97.1 (T3310046C)			1.31
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature above qualifying at 19.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW House	T3310046D	Electrical Conductivity (EC)	EPA 120.1 and	0.228	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	94.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	9.18	mg/L
		Sodium (Na)	EPA 200.7	15.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	9.73	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.87	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	135	mg/L
		Calcium (Ca)	EPA 200.7	17.0	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	4.81	mg/L
		Soluble Salts (SALT-SOL)	Calculation	146	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)			3.82
Calcium (Ca)	92.1	99.5 (T3310046C)	99.9 (T3310046C)			0.329
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	101	94.4 (T3310046C)	95.2 (T3310046C)			0.810
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	92.0	95.8 (T3310046C)	97.1 (T3310046C)			1.31
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation /Ground Water (H14)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Alm 6	T3310046E	Electrical Conductivity (EC)	EPA 120.1 and	0.233	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	94.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	7.41	mg/L
		Sodium (Na)	EPA 200.7	15.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	9.71	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.77	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	145	mg/L
		Calcium (Ca)	EPA 200.7	17.1	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	2.37	mg/L
		Soluble Salts (SALT-SOL)	Calculation	149	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)			3.82
Calcium (Ca)	92.1	99.5 (T3310046C)	99.9 (T3310046C)			0.329
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	101	94.4 (T3310046C)	95.2 (T3310046C)			0.810
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	92.0	95.8 (T3310046C)	97.1 (T3310046C)			1.31
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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Compliance Analysis Report

Customer
Martin Land & Cattle
7532 S. Henry Rd
Farmington, CA 95230

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation /Ground Water (H14)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 20.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Alm 13	T3310046F	Electrical Conductivity (EC)	EPA 120.1 and	0.233	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	96.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	8.37	mg/L
		Sodium (Na)	EPA 200.7	14.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	9.58	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.72	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	150	mg/L
		Calcium (Ca)	EPA 200.7	16.9	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	4.03	mg/L
		Soluble Salts (SALT-SOL)	Calculation	149	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)			3.82
Calcium (Ca)	92.1	99.5 (T3310046C)	99.9 (T3310046C)			0.329
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	101	94.4 (T3310046C)	95.2 (T3310046C)			0.810
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	92.0	95.8 (T3310046C)	97.1 (T3310046C)			1.31
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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Grower

Consultant:
F & R Ag Services
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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/6/23 2:15 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/7/23

Irrigation /Ground Water (H14)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 20.0

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Walnut	T3310046G	Electrical Conductivity (EC)	EPA 120.1 and	0.231	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	96.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	8.28	mg/L
		Sodium (Na)	EPA 200.7	15.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	9.65	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.74	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	140	mg/L
		Calcium (Ca)	EPA 200.7	17.1	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	3.24	mg/L
		Soluble Salts (SALT-SOL)	Calculation	148	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	92.2 (T3310046A)	95.8 (T3310046A)			3.82
Calcium (Ca)	92.1	99.5 (T3310046C)	99.9 (T3310046C)			0.329
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	101	94.4 (T3310046C)	95.2 (T3310046C)			0.810
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	92.0	95.8 (T3310046C)	97.1 (T3310046C)			1.31
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Martin Land & Cattle
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Joe
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/6/23		DW A	Domestic	18.8		X				
2			↓ B	↓	20.8		✓				
3		mLC	IW 1	Irrigation	21.2						H14
4			DW House		19.2						
5			IW Alm 6		18.4						
6			IW Alm 13		20.2						
7			IW Walnut		20.0						
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
	Jeanette Sewell	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	11-6-23 2115			T3310046

(LABORATORY USE ONLY)

2023 Annual Report- Martin Land & Cattle

Final Audit Report 2024-06-25

Created:	2024-06-21
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAwWBSaD0JeUdmDv5sT4cgPVTx4A4ps4FD

"2023 Annual Report- Martin Land & Cattle" History

Document created by Kristen Jones (kjones@fragservices.com)
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Signer martinjoe6100@aol.com entered name at signing as joe martin
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